

SURVEY

Incremental Learning in Medical Imaging: A Comprehensive Survey of Deep Neural Network Advances and Challenges

FATMA HARMAN^{ID}, (Associate Member, IEEE)

Department of Electrical and Electronics Engineering, Manisa Celal Bayar University, 45140 Manisa, Türkiye

e-mail: fatmaharman89@gmail.com

ABSTRACT Deep neural networks (DNNs) have achieved remarkable success in medical image analysis; however, their deployment in real-world clinical environments remains constrained by the static nature of conventional training pipelines. In practice, medical data evolve continuously as new imaging modalities, institutions, and disease categories emerge, necessitating adaptive models capable of learning incrementally without catastrophic forgetting. While lifelong learning (LL) and continual learning (CL) frameworks broadly target continuous model adaptation, incremental learning (IL) emphasizes structured, phase-wise knowledge acquisition involving new classes, domains, or modalities. Despite increasing attention to CL in computer vision, IL-specific methodologies, evaluation protocols, and benchmarks tailored to medical imaging remain largely underexplored. This paper presents the first systematic survey dedicated to incremental learning in medical imaging. A unified taxonomy is proposed across three complementary axes: i) *Learning Mechanisms* — including regularization-, replay-, dynamic architecture-, parameter modulation-, and hybrid/meta-based methods; ii) *Incremental Scenarios* — covering class-, domain-, task-, modality-, and data/online-incremental settings; and iii) *Application Dimensions* — encompassing task types, imaging modalities, data-access constraints, and challenge-driven factors such as imbalance, forgetting, and privacy. For each axis, representative algorithms, evaluation protocols, and benchmark datasets are synthesized to illustrate how IL strategies extend continual learning principles toward clinically deployable AI systems. Finally, the survey highlights emerging research directions—such as privacy-preserving and federated IL, data-free adaptation, uncertainty calibration, and scalable integration with foundation models—laying the groundwork for robust and sustainable clinical AI development.

INDEX TERMS Incremental learning, continual learning, lifelong learning, medical imaging, catastrophic forgetting, deep neural networks, domain adaptation, privacy-preserving AI, federated learning, foundation models.

I. INTRODUCTION

Deep learning (DL) has become a cornerstone of modern medical image analysis, achieving expert-level performance in classification, segmentation, and detection tasks across modalities such as MRI, CT, and ultrasound [1], [2]. However, most DL models rely on static, centralized, and fully labeled datasets, overlooking the dynamic and evolving nature of real-world healthcare data. In clinical practice,

data distributions change continuously due to variations in scanners, imaging protocols, demographics, and emerging diseases [3], [4], [5], [6]. For instance, a DL system developed by Google for diabetic retinopathy screening achieved high offline accuracy but experienced severe performance degradation after deployment [5]. Such domain shifts reveal the fragility of static learning paradigms and underscore the need for adaptive models capable of continuous knowledge integration.

Traditional approaches such as transfer learning and domain adaptation partially mitigate these issues by reusing

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learned representations or aligning feature distributions across domains [6], [7], [8]. Yet, they often assume joint access to source and target data—an unrealistic expectation in privacy-sensitive medical contexts where datasets are decentralized and heterogeneous. Moreover, fine-tuning on new data commonly induces *catastrophic forgetting*, where performance on previously learned tasks deteriorates sharply [9], [10]. These challenges have spurred growing interest in *lifelong*, *continual*, and *incremental learning* paradigms that enable progressive adaptation while retaining prior knowledge [11], [12], [13], [14], [15], [16].

Biologically, continual learning draws inspiration from the human brain's ability to balance **stability**—the retention of old knowledge—and **plasticity**—the acquisition of new information [13], [17]. This stability–plasticity dilemma remains central to designing artificial systems capable of long-term learning. While human cognition consolidates relevant memories to adapt to evolving environments, deep neural networks tend to overwrite previously acquired parameters during gradient-based updates. Achieving this trade-off computationally remains one of the key challenges in continual and incremental learning.

In medical imaging, continual and incremental learning are particularly relevant due to rapid technological evolution, inter-institutional variability, and strict privacy regulations [6], [11]. Retraining large-scale models from scratch for each new cohort or device upgrade is computationally infeasible and often restricted by data governance policies. Consequently, *incremental learning (IL)*—a structured form of continual learning where new classes, domains, or tasks are introduced in discrete stages—offers a promising pathway toward scalable and privacy-aware medical AI systems [18], [19], [20], [21].

Recent research exemplifies this transition. Perkonig et al. [22] proposed a dynamic memory framework to mitigate domain shifts in MRI and CT data. Singh et al. [23] demonstrated class-incremental adaptation for multi-modality healthcare models. Chen et al. [24] introduced a classifier-free incremental segmentation framework with contrastive regularization and knowledge distillation. Ayromlou [25] developed complementary frameworks addressing sequential and distributed heterogeneity: (i) a data-free class-incremental pipeline that synthesizes pseudo-samples to reduce forgetting, and (ii) a federated incremental approach that generates global synthetic data to preserve privacy across institutions. Together, these efforts signify a paradigm shift from static learning toward adaptive, distributed, and continuously evolving medical AI [6], [11].

Although continual learning has been widely studied in computer vision and natural language domains, to the best of our knowledge, **no comprehensive survey has been devoted to incremental learning in medical imaging**. Despite recent progress, there is still a lack of unified synthesis connecting the mechanisms, scenarios, and clinical applications of incremental learning. This survey fills this gap by providing

a structured overview of *Incremental Learning in Medical Imaging (ILMI)* along three complementary dimensions:

- 1) **Learning Mechanisms** — regularization-, replay-, architecture-, and hybrid/meta-based strategies to mitigate catastrophic forgetting;
- 2) **Incremental Scenarios** — class-, domain-, task-, modality-, and online-incremental settings reflecting realistic clinical evolution;
- 3) **Application Dimensions** — imaging modalities, data-access constraints, and clinical challenges such as privacy, imbalance, and uncertainty.

Finally, the paper identifies key research challenges and emerging trends, including **data-free incremental learning** [25], **uncertainty calibration** [26], **foundation model adaptation** [24], [27], and **federated continual frameworks** for privacy-sensitive environments. By consolidating methodological and clinical perspectives, this survey provides a roadmap for developing reliable, adaptive, and ethically deployable incremental learning systems in medical imaging.

Paper Organization— The remainder of this paper is organized as follows. Section II outlines the background and problem formulation. Section III presents the proposed taxonomy covering learning mechanisms, incremental scenarios, application dimensions, and the scope and limitations of incremental learning in medical imaging. Section IV reviews evaluation protocols, evaluation metrics, and benchmark datasets commonly used in ILMI studies. Section V discusses open challenges and research trends, while Section VI highlights future perspectives. Finally, Section VII concludes the paper with a summary of key findings and implications for clinical deployment.

A. MOTIVATION AND UNIQUENESS OF THIS SURVEY

The rapid expansion of medical imaging data and continuous evolution of acquisition technologies have rendered traditional static deep learning pipelines inadequate for sustainable clinical deployment (Fig. 1). In real-world healthcare settings, new diseases, imaging devices, and demographic variations constantly shift data distributions over time. Retraining models from scratch or maintaining large historical datasets is computationally expensive and often restricted by privacy and regulatory barriers. Therefore, medical AI systems require adaptive frameworks capable of integrating new knowledge incrementally without catastrophic forgetting.

Although *lifelong learning (LL)*, *continual learning (CL)*, and *incremental learning (IL)* are often used interchangeably, they differ in scope and granularity. *Lifelong learning* broadly refers to continuous accumulation and transfer of knowledge across diverse tasks and domains throughout a model's operational lifespan [19], [28]. *Continual learning* focuses on enabling models to adapt to non-stationary data streams, often without explicit task boundaries [12], [17]. In contrast, *incremental learning* represents a structured subset of CL, where

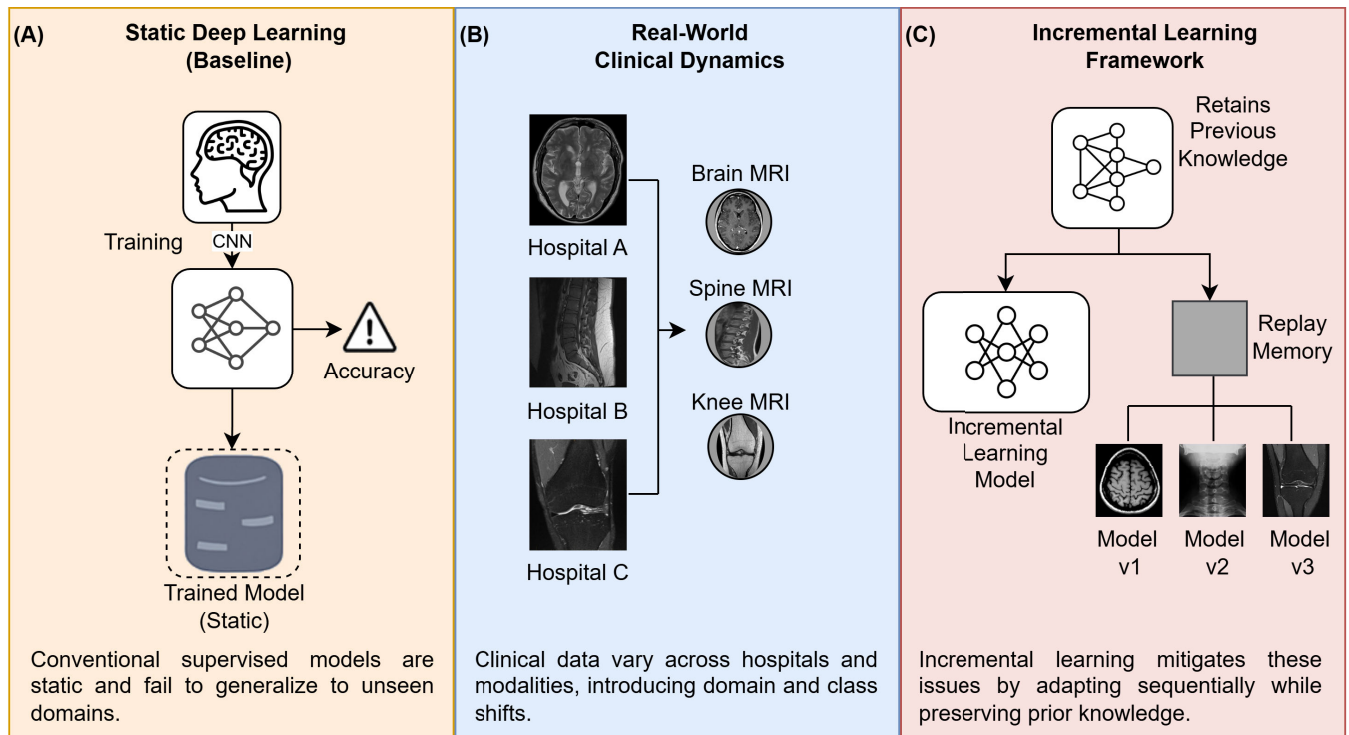


FIGURE 1. Illustration of the transition from static deep learning to incremental learning in medical imaging. (A) Conventional supervised models are static and fail to generalize to unseen domains. (B) Clinical data vary across hospitals and modalities, introducing domain and class shifts. (C) Incremental learning mitigates these issues by adapting sequentially while preserving prior knowledge.

new knowledge—such as classes, domains, or modalities—is introduced in discrete stages [6], [20]. Within medical imaging, IL serves as a pragmatic and clinically interpretable formulation of CL, supporting controlled model updates under privacy and data governance constraints. This makes IL particularly well-suited for cross-hospital adaptation, multi-modality integration, and longitudinal patient monitoring.

Despite growing interest, research on IL in medical imaging remains fragmented across learning paradigms, modalities, and task types. Existing surveys have primarily addressed general CL algorithms [17], [19] or focused on specific medical tasks such as segmentation or classification [11], [13]. To the best of our knowledge, no prior work systematically examines *incremental learning* as a distinct paradigm within medical imaging. While several reviews discuss CL in general or isolated clinical contexts, none provide an integrative synthesis connecting learning mechanisms, incremental scenarios, and application dimensions under a unified taxonomy tailored to the challenges of medical data. Accordingly, this paper presents the first comprehensive survey devoted exclusively to incremental learning in medical imaging.

Uniqueness. This work distinguishes itself through the following key contributions:

- **Unified Taxonomy:** A structured categorization of incremental learning mechanisms (regularization-, replay-, dynamic-, and architecture-based) contextualized for medical imaging.

- **Scenario-Driven Framework:** A systematic organization of incremental scenarios (class-, domain-, task-, and modality-incremental) reflecting realistic clinical evolution.
- **Comprehensive Comparative Analysis:** Integration of application dimensions, including imaging modality, task type, data accessibility, and privacy considerations.
- **Emerging Research Directions:** Discussion of open challenges such as data-free incremental learning, uncertainty calibration [26], foundation model adaptation [8], [24], and federated continual frameworks.

Collectively, these contributions position this survey as a comprehensive roadmap for developing adaptive, privacy-preserving, and clinically generalizable incremental learning systems in medical imaging.

II. BACKGROUND AND PROBLEM FORMULATION

A. CONCEPTUAL FOUNDATIONS

Continual, *incremental*, and *lifelong learning* have emerged as central paradigms for building adaptive deep learning systems capable of handling non-stationary data. Although these terms are sometimes used interchangeably, each represents a distinct scope of model adaptation. *Continual learning (CL)* refers to a model's ability to update its knowledge as new data arrive over time, often without explicit task boundaries. *Lifelong learning (LL)* extends this idea toward open-ended, cumulative knowledge acquisition across longer time horizons, emphasizing sustained reasoning and knowledge

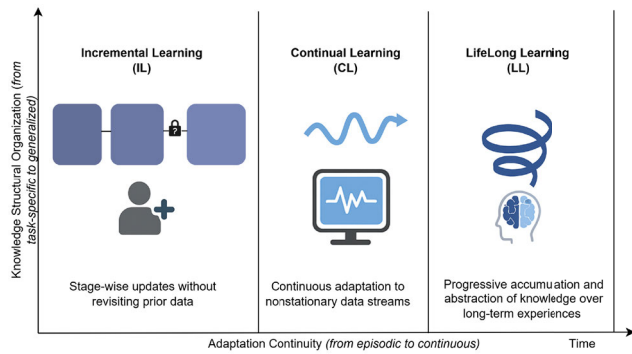


FIGURE 2. Conceptual relationships and distinctions among incremental, Continual, and Lifelong learning paradigms. Incremental learning performs stage-wise updates without revisiting prior data; continual learning adapts continuously to non-stationary data streams; and lifelong learning accumulates and abstracts knowledge across long-term experiences. Axes represent the continuum from episodic to continuous adaptation (x-axis) and from task-specific to generalized knowledge organization (y-axis).

transfer. In contrast, *incremental learning (IL)* operates in a structured, phase-based manner, where new tasks, classes, or domains are introduced sequentially in discrete stages. The model must learn new information while preserving prior performance—an especially relevant scenario for medical imaging, where new diseases, modalities, or scanner types emerge periodically rather than continuously.

A concise comparison of CL, IL, and LL paradigms is presented in Table 1. The table highlights their conceptual overlaps and distinctions, showing why IL serves as a practical bridge between continuous adaptation and long-term knowledge retention. Specifically, CL enables learning from non-stationary input streams, IL formalizes this process into discrete stages without revisiting prior data, and LL extends it toward persistent, cross-task accumulation of expertise.

Together, these paradigms illustrate how IL inherits the stability–plasticity balance of CL while maintaining the structured scalability and generalization goals of LL—features that are essential for real-world clinical environments. Figure 2 visualizes these conceptual relationships, positioning CL, IL, and LL along a continuum from episodic to continuous adaptation and from task-specific to generalized knowledge organization.

B. THE CATASTROPHIC FORGETTING PROBLEM

The primary challenge in both continual and incremental learning is *catastrophic forgetting*—the abrupt loss of previously acquired knowledge when adapting to new tasks. During gradient-based optimization, parameters that encode old information are often overwritten by updates from new data, leading to severe performance degradation on previously learned classes or tasks. This phenomenon reflects the fundamental *stability–plasticity dilemma*: a trade-off between the ability to preserve prior knowledge (*stability*) and to acquire new information (*plasticity*) [17]. While the human brain achieves this balance through mechanisms such

as hippocampal replay and synaptic consolidation, artificial neural networks lack intrinsic memory consolidation, making them highly vulnerable to forgetting.

To address this issue, several families of algorithmic strategies have been proposed. **Regularization-based methods** constrain parameter updates to protect weights important for prior tasks—identified via Fisher information or gradient sensitivity—to prevent drastic shifts from previously learned representations (e.g., Elastic Weight Consolidation or Synaptic Intelligence). **Rehearsal-based methods** mitigate forgetting by storing or synthesizing representative samples from earlier tasks and interleaving them with new data during training, effectively approximating inter-task memory replay. **Dynamic architectural methods** allocate new subnetworks or expand existing ones to accommodate new tasks while maintaining previously learned functionality.

Hybrid approaches combine these paradigms with techniques such as meta-learning, generative replay, or attention modulation to jointly optimize knowledge retention and adaptability. These strategies aim to maintain long-term performance while enabling flexible integration of new information across evolving task distributions.

In medical imaging, catastrophic forgetting frequently arises when a model fine-tuned on new hospital datasets, imaging modalities, or disease cohorts loses diagnostic accuracy on earlier domains. This problem is exacerbated by institutional data silos and strict privacy regulations that restrict access to historical samples, rendering replay or joint training infeasible. Thus, mitigating catastrophic forgetting is not merely a technical objective but a clinical necessity for ensuring safe, reliable, and continually adaptive medical AI systems.

C. PROBLEM FORMULATION

Formally, the incremental learning (IL) problem can be defined as a sequence of tasks $\{T_1, T_2, \dots, T_N\}$, where each task T_i is associated with a dataset $D_i = \{(x, y)\}$. During the training of task T_i , the model has access only to the current dataset D_i , while previous datasets $\{D_1, \dots, D_{i-1}\}$ are unavailable due to privacy, memory, or regulatory constraints. The objective is to learn a model f_θ that maintains strong performance across all tasks $\{T_1, \dots, T_i\}$ after each incremental stage, without suffering from catastrophic forgetting. This objective can be expressed as:

$$\max_{\theta_i} \mathbb{E}_{(x,y) \in D_{\leq i}} [\mathcal{L}(f_{\theta_i}(x), y)], \quad (1)$$

where θ_i denotes the model parameters after learning task T_i , and $D_{\leq i}$ represents the (implicit) cumulative data distribution of all tasks up to stage i . The optimization is constrained such that θ_i is updated incrementally without full access to historical datasets, that is, $\theta_i = \mathcal{A}(\theta_{i-1}, D_i)$, where $\mathcal{A}$ denotes the adaptation operator governing the learning process.

Depending on what varies across consecutive tasks—such as class labels, imaging domains, or task objectives—the IL setting can be categorized as *class-incremental*,

TABLE 1. Conceptual comparison among continual, incremental, and lifelong learning paradigms.

Aspect	Continual Learning (CL)	Incremental Learning (IL)	Lifelong Learning (LL)
Data Stream	Continuous, non-stationary input	Sequential, discrete training stages	Open-ended, cumulative learning across time
Task Boundaries	Often undefined or fuzzy	Explicit and well-defined	Broad, task-agnostic with cumulative reasoning
Memory Mechanism	Regularization or replay-based	Rehearsal, distillation, or architecture growth	Knowledge base, transfer, or meta-learning
Goal	Adaptation without forgetting	Retention and plastic adaptation across stages	Long-term knowledge transfer and consolidation
Typical Medical Analogy	Adapting to evolving scanner protocols or acquisition settings	Integrating new disease classes or imaging modalities	Building a generalized diagnostic model over time

domain-incremental, or *task-incremental* [19], [20]. These canonical formulations define the degree of task relatedness and expected knowledge transfer between stages, forming the foundation for evaluation protocols and algorithmic taxonomies discussed in later sections.

Beyond these canonical formulations, recent extensions introduce more dynamic paradigms such as *modality-incremental learning* (adapting to new imaging modalities) and *online or streaming-incremental learning* (adapting continuously to incoming data). These extensions address realistic medical scenarios where data evolve asynchronously across institutions, modalities, and time—a topic further detailed in Section III-C.

D. CHALLENGES IN MEDICAL IMAGING

Applying incremental learning in medical imaging introduces several unique challenges in contrast with natural image domains. These challenges stem from data governance, annotation requirements, and the inherent heterogeneity of clinical data sources:

- **Privacy and Governance:** Centralized data storage is often restricted under strict privacy regulations (e.g., HIPAA, GDPR), limiting the feasibility of replay or joint training strategies.
- **Annotation Scarcity:** Expert annotations from radiologists or pathologists are expensive and time-consuming to obtain, hindering large-scale or frequent incremental updates.
- **Domain Heterogeneity:** Variations in scanner vendors, acquisition protocols, and patient demographics introduce substantial domain shifts, even within the same imaging modality.
- **Data Imbalance and Rare Diseases:** Newly introduced classes often correspond to rare conditions, exacerbating model bias and accelerating forgetting.
- **Ethical and Clinical Safety:** Maintaining diagnostic performance on previously learned diseases is essential; catastrophic forgetting could lead to clinically harmful outcomes.

These factors make incremental learning in medical imaging not only a technical challenge but also a matter of clinical reliability and ethical responsibility, emphasizing the need

for privacy-preserving, data-efficient, and domain-adaptive IL frameworks.

E. RELATIONSHIP TO RELATED PARADIGMS

Incremental learning (IL) shares conceptual foundations with several adjacent paradigms in machine learning but differs in its goals, assumptions, and operational constraints. The following summarizes the key distinctions:

- **Transfer Learning:** Reuses pretrained representations to accelerate convergence on new tasks but typically assumes joint access to both source and target data. This assumption is often unrealistic in decentralized, privacy-sensitive medical environments.
- **Domain Adaptation:** Seeks to align feature distributions between source and target domains to improve cross-domain generalization. However, it does not explicitly address sequential task learning or long-term knowledge retention.
- **Open-World Learning:** Focuses on the continuous discovery and recognition of novel classes in unconstrained environments but generally lacks the explicit memory consolidation and stability mechanisms that characterize incremental learning.

While closely related to continual and lifelong learning, incremental learning differs by its structured, stage-wise formulation, where new knowledge—such as classes, domains, or modalities—is introduced sequentially under strict data access constraints. Moreover, incremental learning naturally complements federated and privacy-preserving frameworks, where data remain distributed across institutions. Similarly, unlike open-set recognition or domain generalization, it explicitly models temporal task evolution rather than single-step domain shifts.

In summary, incremental learning provides a structured and privacy-aware framework for progressive model adaptation to evolving clinical data, effectively bridging the gap between static learning and fully dynamic lifelong systems.

F. SUMMARY AND CONCEPTUAL ILLUSTRATION

The evolution from static deep learning toward adaptive and incremental paradigms represents a pivotal transition in the pursuit of clinically deployable AI systems. Conventional

supervised models are typically trained once on fixed datasets and remain static thereafter, limiting their ability to generalize to new domains, institutions, or imaging protocols. As depicted in Fig. 1, real-world clinical environments are inherently dynamic, characterized by a continuous influx of data from diverse scanners, hospitals, and patient populations. Under such distributional shifts, static models often experience severe performance degradation, whereas incremental learning frameworks enable sequential adaptation by integrating new information—such as diseases, modalities, or patient cohorts—without retraining from scratch.

At a broader conceptual level, Fig. 2 positions incremental learning within the continuum of continual and lifelong learning paradigms. *Continual learning (CL)* focuses on adapting to non-stationary data streams through ongoing parameter updates, while *incremental learning (IL)* operates in a stage-wise manner, updating models with new tasks or domains without direct access to previous data. In contrast, *lifelong learning (LL)* extends beyond short-term adaptability, emphasizing cumulative knowledge abstraction and transfer across long temporal scales.

Collectively, these paradigms define a spectrum of adaptability and retention, wherein IL serves as a pragmatic bridge—combining the continuous adaptability of CL with the long-term consolidation objectives of LL. This balance is particularly valuable in medical imaging, where data evolve gradually yet persistently across time, institutions, and technologies. Together, Figs. 1 and 2 illustrate both the practical motivation and conceptual positioning of incremental learning within adaptive medical AI. From a clinical perspective, such adaptability enables AI systems to remain reliable across evolving scanners, populations, and disease landscapes—an essential step toward long-term clinical integration.

The following section formalizes the incremental learning problem and describes its main variants across class, domain, task, and modality dimensions.

III. TAXONOMY OF INCREMENTAL LEARNING IN MEDICAL IMAGING

Incremental learning in medical imaging encompasses diverse algorithmic strategies and clinical contexts that enable models to adapt to evolving data distributions without complete retraining or access to historical data. To provide a structured overview of this rapidly expanding field, existing approaches are categorized along three orthogonal axes, as illustrated in Fig. 3: (1) *Learning Mechanisms* — the algorithmic principles that support knowledge retention and adaptation; (2) *Incremental Scenarios* — the structural forms of data and task evolution; and (3) *Application Dimensions* — task- and modality-specific contexts encompassing data accessibility, privacy, and clinical constraints.

Together, these axes constitute a comprehensive taxonomy that unifies methodological advances with practical deployment needs in medical AI. This organization also facilitates cross-comparison of incremental learning algorithms

by clarifying how different strategies address catastrophic forgetting, domain shifts, and the stability–plasticity trade-off inherent in evolving medical data.

A. LEARNING MECHANISMS

The core principle of incremental learning lies in mitigating *catastrophic forgetting* while continuously integrating new information into existing representations. Based on their underlying strategies, incremental learning approaches can be broadly categorized into five main families:

- 1) Regularization-based methods,
- 2) Replay-based methods,
- 3) Dynamic and architecture-adaptive methods,
- 4) Selective parameter modulation methods, and
- 5) Hybrid and meta-learning-based methods.

Each family employs distinct mechanisms to balance the *stability–plasticity* trade-off during incremental adaptation. Although these mechanisms originate largely from general continual learning literature, their behavior in medical imaging may differ substantially due to (i) limited and costly expert annotations, (ii) class imbalance and long-tailed pathology distributions, (iii) domain shift across scanners, institutions, and acquisition protocols, (iv) 3D volumetric and multimodal data streams (e.g., MRI/CT/US combined with clinical metadata), and (v) strict privacy and governance constraints that often limit data replay or centralized retraining. Consequently, beyond describing core algorithmic principles, we also emphasize the practical suitability of each learning mechanism family under clinically realistic incremental scenarios, including privacy-preserving adaptation and robustness under longitudinal distribution drift. A structured view of how incremental learning manifests across different *medical imaging task types* (e.g., classification, segmentation, detection) and *modalities* is further provided in Section III-D and summarized in Table 4.

Table 2 summarizes the major learning mechanism families and highlights their representative strategies and characteristics, along with a brief medical-imaging-oriented suitability note.

The subsequent subsections elaborate on each family of learning mechanisms, emphasizing both their adaptation principles and representative applications in medical imaging.

1) REGULARIZATION-BASED METHODS

Regularization-based approaches in continual and incremental learning (CL/IL) aim to mitigate *catastrophic forgetting* by constraining parameter updates on weights crucial for previously learned tasks. The central idea is to introduce an additional penalty term in the loss function that discourages large deviations in important parameters, thereby balancing the trade-off between *stability* (retaining prior knowledge) and *plasticity* (acquiring new knowledge).

Early foundational methods such as Elastic Weight Consolidation (EWC) [9], Synaptic Intelligence (SI) [29],

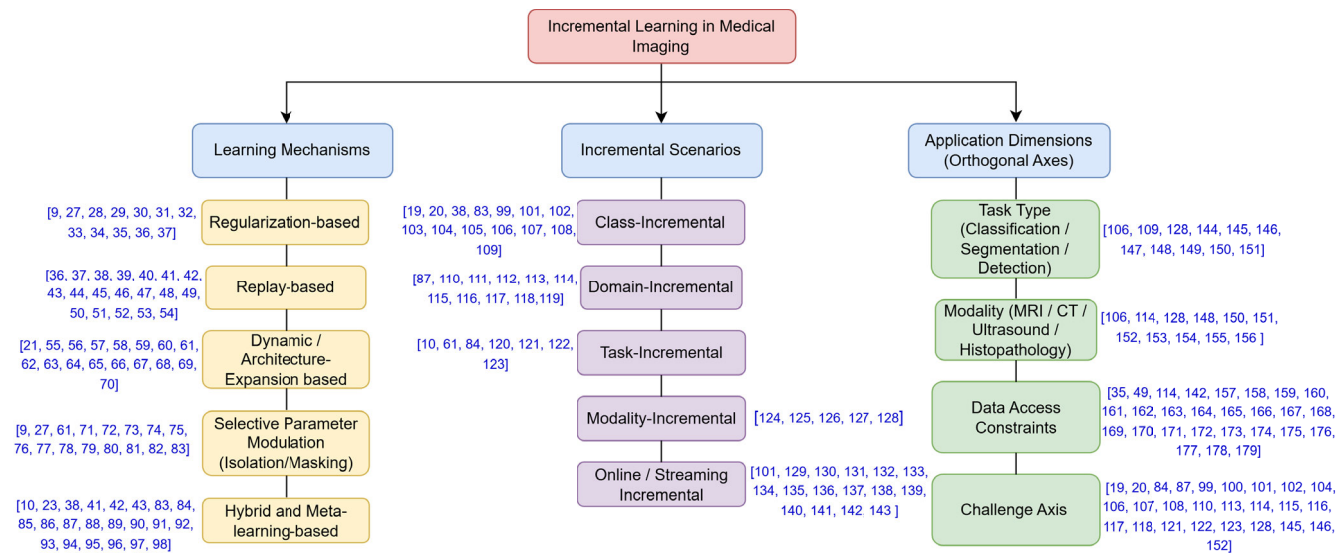


FIGURE 3. Illustration of the proposed taxonomy for incremental learning in medical imaging. The taxonomy delineates three orthogonal axes—Learning mechanisms, Incremental scenarios, and Application dimensions—that together describe how models adapt to evolving data distributions while addressing catastrophic forgetting, domain shifts, and clinical constraints.

TABLE 2. Comparison of major learning mechanism families in incremental learning, highlighting their core principles, computational and memory characteristics, and representative algorithms.

Aspect	Regularization-based	Replay-based	Dynamic-based	Architecture Expansion-based	Parameter Isolation-based	Parameter Masking-based	Hybrid-based	Meta-based
Core Idea	Constrain weight updates to retain previous knowledge.	Reuse stored or generated samples to mitigate forgetting.	Adapt network routing or structure dynamically without permanent expansion.	Expand model capacity by adding neurons, filters, or modules for new tasks.	Assign disjoint parameter subsets to isolate task knowledge.	Learn binary or soft masks to activate task-specific parameters.	Combine multiple strategies (e.g., replay + regularization or expansion + masking) for balanced plasticity.	Use meta-learning to optimize rapid adaptation and transfer across tasks.
Memory Requirement	Low	Medium–High	Medium	High (grows with tasks)	Medium	Low–Medium	Medium	Low–Medium
Computation Overhead	Low	Moderate	Moderate–High	High	Moderate	Moderate	High	High
Task-ID Dependency	No	No	Often Yes	Yes	Yes	Often Yes	Depends on combination	No
Forgetting Resistance	Moderate	High	High	High	High	High	Very High (multi-mechanism synergy)	Moderate–High
Scalability	High	Medium	Medium	Low–Medium	Medium	High	Medium	High
Representative Methods	EWC, LwF, SI, MAS	iCaRL, ER, DGR, GDumb	DEN, RCL, PathNet	ProgressiveNet, ExpertGate, Net2Net	PackNet, Piggyback, PathNet variants	HAT, SupSup, IMM	LUCIR, IL2M, DER, RATT	ANML, Meta-Experience Replay, OML, MAML-based CL
Typical Use Case	Lightweight continual classification with minimal resources.	Class-incremental or online setups leveraging memory or generative replay.	Task-adaptive routing and multi-domain incremental setups.	Task-incremental or multi-task learning requiring modular capacity growth.	Modular continual learning with explicit weight partitioning.	Compact continual learners using selective activation.	Balancing stability–plasticity via hybrid strategies.	Fast adaptation and transfer in dynamic environments.
Medical suitability / notes	Privacy-friendly; may under-adapt under severe scanner/domain drift.	Strong retention; constrained by privacy, storage, and synthetic realism.	Suitable for multi-center drift; higher complexity/compute.	Useful for new modalities/tasks; capacity grows with tasks.	Stable long-term; often requires task identity and careful design.	Parameter-efficient; task-ID often needed, risk of under-sharing.	Strong stability–plasticity; tuning and complexity may increase.	Promising for few-shot/low-label shifts; limited clinical validation so far.

and Memory Aware Synapses (MAS) [30] formalized this concept through importance-weighted regularization. These approaches estimate parameter importance using Fisher information (EWC), online relevance accumulation (SI), or sensitivity-based measures (MAS). Later

extensions—including KFAC-EWC [31] and Riemannian Walk (RWalk) [32]—refined this framework by modeling curvature in the loss landscape and incorporating geometric constraints on optimization trajectories. Subsequent variants introduced uncertainty-aware penalties [33], [34],

adjusting regularization strength according to Bayesian confidence or self-attention-derived uncertainty, and node-importance-based adaptive sparsity [35], which dynamically freezes or reinitializes parameters based on relevance estimates. Conceptor-aided backpropagation (CAB) [36] further extended these ideas via nonlinear constraints that preserve previously learned representations.

Although most of these algorithms were validated on general-purpose benchmarks such as MNIST, CIFAR, and ImageNet, they established the theoretical foundation for *data-free continual learning*—a particularly relevant property for medical imaging, where privacy regulations often prohibit data replay or centralized retraining.

Building upon these foundations, recent studies have explored regularization-based continual learning in medical imaging contexts. Verma et al. [37] conducted a comparative analysis of privacy-preserving CL algorithms for optical coherence tomography (OCT) and colon histopathology classification, showing that non-replay regularization methods can mitigate forgetting while maintaining patient confidentiality. Yavari and Furst [38] combined knowledge distillation with penalty-based constraints for incremental prostate MRI analysis on the PI-CAI dataset, achieving robust cross-hospital knowledge retention without data sharing. Zhang et al. [39] extended the regularization paradigm to multimodal biomedical imaging by integrating contrastive regularization with low-rank adaptation (LoRA), enabling cross-modality generalization.

Overall, these studies demonstrate that regularization-based continual learning represents a practical and privacy-compliant framework for developing adaptive medical AI systems capable of maintaining diagnostic reliability across evolving clinical settings.

2) REPLAY-BASED METHODS

Replay-based methods aim to mitigate *catastrophic forgetting* by explicitly *replaying or regenerating past knowledge* through the reintroduction of previous data or their approximations. In contrast to regularization-based strategies that constrain parameter updates indirectly, replay methods directly recall prior experiences at either the *sample* or *feature* level—by maintaining a compact exemplar buffer (sample replay) or synthesizing past distributions using generative models (generative replay).

Early foundational studies established the basis of this paradigm. The iCaRL framework [40] combined exemplar-based memory with knowledge distillation for joint feature and classifier learning. Gradient Episodic Memory (GEM) and its variant A-GEM [41] introduced gradient projection to ensure backward compatibility of updates, balancing stability and plasticity. Deep Generative Replay (DGR) [42] trained generative adversarial networks (GANs) to synthesize pseudo-samples representing previous tasks, eliminating the need to store raw data. Subsequent studies refined these ideas through class-imbalance correction [43],

feature-space replay via representation distillation [44], [45], and hybrid memory schemes that combined real and generated exemplars [46]. Prototype-similarity replay unified prototype alignment and regularization in a shared embedding space [47], while adaptive replay scheduling optimized rehearsal frequency and sample selection [48].

Applications in Medical Imaging:—Replay-based learning has become particularly relevant in medical imaging, where privacy regulations and domain heterogeneity limit centralized data access. Within the *generative replay* line, GAN- and diffusion-based models have been adopted to retain task knowledge under strict data retention constraints. For example, generative replay improved longitudinal stability in cardiac MRI and retinal fundus analysis while maintaining compact memory footprints [49]. SegGAN introduced joint image–mask synthesis with pseudo-label filtering for incremental medical segmentation [50], and conditional diffusion replay achieved robust domain-incremental adaptation across diabetic retinopathy and dermoscopic datasets [51].

On the *sample replay* side, selective exemplar strategies prioritized informative yet atypical samples for optimal memory utilization [52]. Dual-stream architectures combining pseudo-label replay with rigidity–plasticity control achieved effective incremental adaptation in heterogeneous brain tumor MRI segmentation [53]. Replay-based active learning (RBACA) integrated exemplar selection into incremental cardiac image classification under limited annotation budgets [54]. In histopathology and retinal imaging, large-scale evaluations confirmed that replay-based continual learning outperforms regularization-only baselines in both performance retention and privacy compliance [37], [55], [56]. Hybrid replay–distillation pipelines also proved effective in multi-center MRI and OCT studies, where synthetic exemplars guided model retention without sharing sensitive data [38]. Similarly, contrastive regularization applied to LoRA modules enabled feature-level replay for multimodal biomedical imaging, enhancing cross-domain transferability while reducing computation [39].

Promising research directions include: (i) advancing *generative and diffusion-based replay* as privacy-preserving surrogates for data sharing; (ii) leveraging *latent-space or prototype-level replay* to handle cross-institutional heterogeneity; and (iii) developing *adaptive exemplar selection and scheduling* strategies for scalable long-term deployments. Despite their effectiveness, replay-based methods still face challenges related to the realism of synthetic data, the computational cost of continual retraining, and memory scalability. Hybrid architectures that integrate replay with distillation and contrastive regularization therefore represent a promising path toward sustainable, privacy-aware incremental learning in medical imaging.

3) DYNAMIC AND ARCHITECTURE EXPANSION-BASED METHODS

A major direction in continual and incremental learning research focuses on expanding or dynamically adapting

model architectures to accommodate new knowledge without overwriting previously learned representations. These approaches can be broadly grouped into two main categories: (i) *architecture expansion-based methods*, which explicitly grow the network to increase representational capacity, and (ii) *dynamic/adaptive methods*, which implicitly reconfigure internal parameters or routing mechanisms without modifying the overall structure. Both paradigms address the stability–plasticity trade-off but differ in how they allocate capacity and preserve prior knowledge.

a: ARCHITECTURE EXPANSION-BASED METHODS

These methods explicitly enlarge the network structure to preserve prior representations while allowing new modules to specialize in novel tasks. The seminal Progressive Neural Network (PNN) [57] introduced lateral connections between task-specific columns to facilitate transfer and prevent forgetting. Building upon this idea, the Dynamically Expandable Network (DEN) [58] enabled selective retraining and unit addition to minimize redundancy during online learning. Li et al. [59] proposed *Learn to Grow*, which adjusts network capacity via task-dependent structural learning, while PackNet [60] utilized iterative pruning and parameter reallocation to accommodate new tasks within a single model. Yan et al. [61] further introduced Dynamically Expandable Representations (DER), augmenting frozen features through mask-based channel addition to balance stability and plasticity. Bio-inspired models have also explored structural plasticity, including neurogenesis-driven expansion [17] and Dynamic Generative Memory [62], which integrate conditional GANs with adaptive masking for capacity growth.

In medical imaging, expansion-based learning has proven effective for modular task integration and continual adaptation. Zhang et al. [63] developed a continual multi-organ segmentation framework that replaces a unified output layer with lightweight class-specific heads, minimizing interference between new and existing structures. Dong et al. [64] proposed ConSept, an adapter-based Vision Transformer (ViT) incorporating lightweight attention adapters for continual semantic segmentation. Similarly, Saeed et al. [65] introduced PEMMA, a parameter-efficient multimodal adaptation framework leveraging LoRA/DoRA modules to extend transformer-based segmentation networks across imaging modalities with minimal overhead. These examples illustrate how architectural modularity and selective expansion promote scalable continual learning in domain-constrained medical environments.

b: DYNAMIC/ADAPTIVE METHODS

Dynamic methods maintain a fixed architecture but adaptively reconfigure parameters, routing paths, or attention mechanisms in response to new data. Early works such as Expert Gate [66] introduced gating autoencoders to dynamically select the most relevant expert for each input, enabling scalable lifelong learning without retraining all

modules. Deep Adaptation Networks [67] reused existing filters through linear combinations to achieve compact cross-domain adaptation, while reinforcement-driven architectures dynamically searched for optimal task-specific subnets [68]. Recent approaches such as Selective Nets [69] leverage foundation models (e.g., DINOv2) for task-free dynamic subnetwork selection, and Task-Attentive Multimodal CL (TAM-CL) [70] extends transformer-based systems through dynamic expansion across vision–language modalities. Bayesian variants [71] further exploit parameter uncertainty to control update magnitude and prevent overfitting during adaptation.

Dynamic adaptation has also been applied to medical imaging. Perkonig et al. [22] introduced a dynamic memory mechanism for continual cardiac and lung analysis, addressing domain shifts via online rehearsal and pseudo-domain discovery. Chee et al. [72] proposed a dynamic architecture with expandable representations to learn new disease classes in ophthalmology and dermatology datasets while preserving previously acquired knowledge. These studies demonstrate that dynamic adaptation—through attention modulation, uncertainty-aware updates, or selective routing—can maintain diagnostic consistency across sequential clinical tasks.

c: SUMMARY

Architecture expansion-based methods emphasize explicit modular growth, whereas dynamic/adaptive methods focus on implicit reconfiguration and parameter reuse. Together, these paradigms enable flexible and resilient continual learning systems, increasingly adopted in medical imaging for incremental segmentation, multimodal fusion, and disease progression modeling. Their convergence—combining modular expansion with dynamic adaptation—represents a promising pathway toward scalable, generalizable, and clinically adaptive next-generation AI frameworks.

4) SELECTIVE PARAMETER MODULATION (ISOLATION/MASKING)

Selective parameter modulation forms a major line of research in continual and incremental learning (CL/IL), aiming to mitigate *catastrophic forgetting* by assigning task-specific or importance-aware subsets of parameters to each learning stage. Instead of storing samples or expanding architectures, these methods regulate which parameters are updated and how they adapt during sequential training. This family broadly includes two paradigms: *parameter isolation*, which structurally separates parameters across tasks, and *parameter masking*, which selectively gates shared parameters based on their relevance.

a: PARAMETER ISOLATION-BASED METHODS

Isolation-based approaches explicitly allocate or preserve disjoint parameter subsets across tasks to prevent interference between previously learned and newly introduced knowledge. Early foundations such as Elastic Weight Consolidation

(EWC) [9] and Synaptic Intelligence (SI) [29] introduced importance-weighted regularization to protect task-relevant parameters, effectively reducing updates on crucial weights. These frameworks—though originally validated on benchmarks like MNIST and Atari—established the theoretical basis for importance-aware continual learning.

Zeng et al. [73] extended this principle through Orthogonal Weight Modification (OWM), constraining gradient updates to directions orthogonal to previously learned subspaces. Yoon et al. [74] proposed Additive Parameter Decomposition (APD), decomposing network parameters into task-shared and task-specific components for scalable and order-robust learning.

In medical imaging, Osman et al. [75] introduced Task-based Parameter Isolation (TBPI) for continual foreground segmentation, where task-specific modules isolate evolving video datasets (CDnet2014, DAVIS2016, SegTrack) to mitigate forgetting across temporal sequences. Zhang et al. [63] designed Parameter-Isolation GNN (PI-GNN) for dynamic graph learning, expanding parameters for new structures while freezing earlier ones—an approach increasingly relevant to evolving medical graph representations. Vicente-Sola et al. [76] proposed Low Interference Feature Extraction Subnetworks (LIFES), replacing task-aware isolation with task-agnostic subnetwork selection and lateral regularization. Collectively, these methods emphasize stability–plasticity balance by protecting task-specific representations rather than relying on data replay.

b: PARAMETER MASKING-BASED METHODS

Masking-based methods follow a complementary strategy by using binary or continuous masks to selectively activate parts of a shared network for each task. Serra et al. [77] introduced Hard Attention to the Task (HAT), one of the earliest binary masking methods that preserve task-relevant units during sequential learning. Mallya et al. [78] extended this concept with Piggyback, learning differentiable binary masks over a fixed pretrained network to adapt it to new tasks without modifying base weights—offering a highly parameter-efficient solution.

Later developments include Supermasks in Superposition (SupSup) [79], which encode thousands of tasks through sparse binary masks, and Winning Subnetworks (WSN) [80], which identify reusable task-specific subnetworks following the lottery ticket hypothesis. Konishi et al. [81] proposed Soft Parameter Gating (SPG), replacing hard binary masks with continuous importance weights to enable partial parameter reuse and smoother knowledge transfer.

Recent works further enhanced mask generation through hypernetworks and subnetwork selection. HyperMask [82] employs a hypernetwork to generate semi-binary masks conditioned on task identity, leveraging the lottery ticket hypothesis for efficient subnetwork selection. Yadav and Bansal [83] developed Exclusive Supermask Subnetwork Training (ExSSNet), enforcing non-overlapping subnetworks

for independent learning trajectories. In medical imaging, Qian et al. [84] introduced the Parameter-level Mask-based Adapter (PMA) for 3D continual learning with vision–language models. PMA integrates learnable masks to fuse 3D and 2D features for sequential organ classification, achieving an effective stability–plasticity trade-off without full retraining.

c: SUMMARY

Selective parameter modulation—through isolation or masking—offers a memory-efficient alternative to replay and architecture expansion methods. These strategies emphasize parameter-space regularization rather than data-space rehearsal, aligning naturally with the privacy and scalability requirements of medical imaging. Isolation-based designs such as TBPI [75] demonstrate feasibility for segmentation and temporal analysis, while PMA [84] and HyperMask [82] highlight how parameter-efficient tuning and multimodal adapters can enhance continual adaptation in clinical contexts. Future directions include integrating isolation and soft-masking into unified, task-agnostic frameworks for scalable continual learning across heterogeneous medical datasets.

5) HYBRID AND META-LEARNING-BASED METHODS

While most continual learning (CL) approaches are designed either as purely discriminative (e.g., distillation- or replay-based) or as dynamic architectural expansion strategies, hybrid and meta-learning-based methods aim to combine the advantages of both paradigms. Hybrid approaches integrate knowledge distillation, replay, or bias correction with adaptive mechanisms, whereas meta-learning seeks to optimize the learning process itself—enabling rapid adaptation to new tasks and reducing forgetting. Although many of these methods were originally proposed in the broader CL literature, their underlying principles—progressive adaptation, knowledge transfer, and plasticity–stability balancing—are directly applicable to incremental learning (IL) in medical imaging, where models must sequentially incorporate new pathologies, modalities, or institutions.

a: HYBRID DISTILLATION-BASED INCREMENTAL LEARNING

Early hybrid frameworks introduced knowledge distillation as a regularization mechanism to mitigate catastrophic forgetting while maintaining representational consistency across incremental tasks. Rebuffi et al. [40] proposed iCaRL, which combines exemplar replay and distillation to jointly learn representations and classifiers in class-incremental settings. Li and Hoiem [10] developed Learning without Forgetting (LwF), which preserves prior knowledge without storing old data by aligning current predictions with previous outputs. Douillard et al. [44] extended this concept with PODNet, introducing spatially pooled distillation and proxy vectors per class for enhanced small-task learning. At a larger scale, Wu et al. [43] addressed classifier bias in

large-scale incremental learning via linear bias correction, while Liu et al. [45] proposed Generative Feature Replay, using feature-level replay with distillation to avoid exemplar storage.

In medical imaging, several studies have adapted these hybrid distillation frameworks to handle non-stationary and institutionally heterogeneous data streams. Li et al. [85] introduced dual distillation and ensemble learning for incremental disease recognition, and Tian et al. [86] proposed multi-scale, multi-task distillation for 3D segmentation with uncertainty-weighted supervision. Chen et al. [24] developed a classifier-free incremental segmentation framework trained via contrastive distillation, while Li et al. [87] presented Progressive Distillation (ProTA) with a Data Fingerprint (FIND) module for corneal microscopy segmentation. Other efforts—such as Fang et al. [88] (cross-view mutual distillation) and Zhang et al. [89] (federated cross-incremental self-supervised learning)—extend hybrid distillation to self-supervised and federated setups. Collectively, these studies demonstrate that hybrid distillation enables robust and privacy-aware incremental adaptation across multi-modal and evolving diagnostic settings.

b: META-LEARNING–BASED INCREMENTAL LEARNING

Meta-learning, or “learning to learn,” focuses on optimizing initialization and update rules so that new tasks require minimal adaptation. This principle has driven the development of task-agnostic and data-efficient continual systems that translate well to incremental medical scenarios. Finn et al. [90] introduced the seminal Model-Agnostic Meta-Learning (MAML), which learns initialization parameters that generalize across tasks with few gradient updates. Nichol et al. [91] proposed Reptile, a first-order approximation that reduces computation while retaining efficiency. Javed and White [92] developed OML, which learns representations explicitly optimized to minimize interference, while Riemer et al. [93]’s Meta-Experience Replay (MER) aligns gradients to stabilize updates. Further, Beaulieu et al. [94] proposed ANML, a neuromodulatory meta-learning framework that adaptively gates neural plasticity, achieving state-of-the-art continual performance. Wang et al. [95] introduced Learning to Prompt (L2P), using learnable prompts to dynamically modulate pretrained representations, eliminating the need for task identity. Tao et al. [96] presented the TOPIC framework for Few-Shot Class-Incremental Learning (FSCIL), employing neural gas topology preservation to stabilize feature evolution under few-shot conditions.

In medical imaging, meta-learning remains relatively underexplored but has recently gained attention for its ability to facilitate rapid and stable adaptation under distributional shifts. Wang et al. [97] proposed a sharpness-aware meta-continual framework for universal medical image registration, enabling cross-modality generalization across brain MRI, abdominal CT, and multi-modal tasks. González et al. [98] developed Lifelong nnU-Net,

a standardized continual segmentation framework incorporating meta-configurable modules for transfer and adaptation. These efforts illustrate the emerging role of meta-learning as a catalyst for incremental medical imaging systems that can rapidly generalize to new domains.

c: HYBRID META-DISTILLATION PARADIGMS (EMERGING TREND)

An emerging research direction integrates meta-learning with hybrid distillation to enhance adaptability, knowledge transfer, and representational retention in continual and incremental learning. Zhu et al. [99] introduced a tri-enhanced distillation framework combining stochastic augmentation, adaptive transfer, and uncertainty-guided fusion for stable domain-continuous segmentation. Li et al. [87] proposed Progressive Task-Guided Adapter Learning (ProTA), while Tian et al. [86] developed multi-scale, multi-task distillation pipelines that preserve fine-grained anatomical features while adapting to new pathologies. Ye et al. [100] presented Dual-Alignment Knowledge Retention, aligning task-specific and global knowledge to mitigate forgetting in continual medical segmentation.

Together, these hybrid meta-distillation frameworks mark a step toward generalizable, foundation-style incremental learning in medical imaging—capable of integrating sequential data, cross-domain shifts, and heterogeneous annotation schemas while maintaining stability, adaptability, and clinical reliability.

B. SCOPE AND LIMITATIONS OF INCREMENTAL LEARNING IN MEDICAL IMAGING

Although incremental learning (IL) offers an attractive paradigm for updating models under evolving data distributions, it is not universally applicable to all medical imaging problems. Understanding its scope and limitations is critical for appropriate deployment. This subsection does not introduce an additional taxonomic axis, but provides a complementary perspective that clarifies when incremental learning is appropriate, insufficient, or better replaced by alternative paradigms in medical imaging.

1) LIMITATIONS OF IL

First, IL methods typically assume that new data arrive in a sequential but semantically related manner. When incoming data are highly heterogeneous, weakly related, or correspond to entirely new anatomical regions or imaging physics, incremental adaptation alone may be insufficient. In such cases, large-scale re-training or transfer learning from task-specific or foundation pretrained models is often more effective.

Second, IL struggles when high-quality labels for new tasks are extremely scarce or absent. While some IL frameworks incorporate few-shot or semi-supervised extensions, scenarios with near-zero supervision are better addressed by

self-supervised pretraining, foundation models, or zero-shot learning approaches.

Third, most IL methods rely on approximate mechanisms (e.g., regularization, replay, or distillation) to preserve past knowledge. Under very long task sequences or severe domain shifts, accumulated approximation errors can lead to gradual performance degradation. This limits the practicality of IL for lifelong deployment spanning many years without occasional re-initialization or consolidation phases.

Finally, IL is not designed to correct systematic biases present in previously learned data (e.g., biased labeling practices or outdated clinical guidelines). If earlier knowledge is itself incorrect or obsolete, incremental preservation may be undesirable, and re-training with curated datasets becomes necessary.

2) SCENARIOS WHERE IL IS WELL SUITED

IL is particularly appropriate for problems where (i) new disease categories or subtypes emerge over time, (ii) scanner or protocol changes introduce gradual domain shift, or (iii) institutions wish to update models without sharing historical patient data due to privacy constraints. Typical examples include continual extension of lesion taxonomies, cross-hospital adaptation of segmentation models, and incremental incorporation of new imaging modalities.

3) SCENARIOS WHERE ALTERNATIVES MAY BE PREFERABLE

Transfer learning or full re-training is more suitable when the target task differs substantially from previous tasks, when large annotated datasets become available at once, or when foundational representations must be redesigned (e.g., transition from 2D natural-image-pretrained backbones to 3D medical foundation models). Similarly, domain generalization or test-time adaptation may be preferable when the primary challenge is robustness to unseen domains rather than sequential task expansion.

C. INCREMENTAL SCENARIOS

In medical imaging, incremental and continual learning (IL/CL) can manifest in multiple ways depending on how new knowledge enters the system—through emerging disease types, evolving imaging domains, novel modalities, or continuous data streams. Building upon the taxonomy introduced by Van de Ven and Tolias [101], six major scenarios can be distinguished: *class-incremental learning (CIL)*, *domain-incremental learning (DIL)*, *task-incremental learning (TIL)*, *modality-incremental learning (MIL)*, *online-incremental learning (OIL)*, and *streaming-incremental learning (SIL)*.

Although this categorization originates from continual learning literature, in medical imaging these scenarios naturally align with the incremental paradigm—where models are updated stage by stage as new data, pathologies, or modalities emerge. In practice, incremental learning serves as an operational subset of continual learning, focusing on discrete adaptation cycles rather than continuous parameter drift.

This alignment enables systematic evaluation of how AI systems evolve over time within the constraints of clinical data governance and privacy.

These paradigms mirror how clinical AI systems evolve in real-world deployments. Models may encounter new disease categories, generalize across scanners and institutions, adapt to multi-modality setups such as MRI–ultrasound fusion, or continuously learn from real-time physiological signals like ECG. Each scenario introduces unique challenges for balancing stability and plasticity while adhering to strict data privacy and accessibility constraints.

Figure 4 provides an intuitive overview of these scenarios using representative medical examples, and Table 3 summarizes their core definitions, evaluation protocols, and characteristic challenges.

Summary: Incremental scenarios define the operational granularity of continual adaptation in medical imaging. Understanding these distinctions clarifies how learning frameworks should be evaluated and deployed in practice—whether handling evolving disease taxonomies, inter-institutional variability, or continuous physiological monitoring. Together, these paradigms provide a conceptual blueprint for designing clinically adaptive and privacy-conscious incremental learning systems.

1) CLASS-INCREMENTAL LEARNING (CIL)

Class-incremental learning (CIL) represents one of the most challenging and clinically realistic forms of incremental adaptation in medical AI. In this setting, new categories are introduced sequentially without explicit task identifiers, requiring the model to integrate novel knowledge while retaining performance on previously learned classes. Although CIL was originally studied within the continual learning (CL) framework, it directly aligns with the incremental paradigm where discrete learning stages correspond to the gradual expansion of disease taxonomies or anatomical labels.

Early foundations such as iCaRL [40]—which combines exemplar replay with joint representation learning—and Progress & Compress [102], which alternates between learning and consolidation using dual network columns, established core principles for incremental adaptation. Van de Ven and Tolias [101] formalized the TIL/DIL/CIL taxonomy and demonstrated that regularization alone is insufficient for true class-incremental learning, emphasizing the necessity of replay and distillation mechanisms. Subsequent systematic analyses by De Lange et al. [19], Masana et al. [20], and Mai et al. [103] benchmarked state-of-the-art methods under varying memory and streaming constraints, consistently favoring hybrid replay–distillation strategies.

In medical imaging, CIL plays a critical role in enabling adaptive disease recognition pipelines and evolving diagnostic taxonomies. Representative examples include dual-distillation and ensemble-based frameworks for incremental disease classification [85], and continual thoracic

TABLE 3. Comparative overview of incremental learning scenarios in medical imaging.

Aspect	CIL	DIL	TIL	MIL	OIL	SIL
Definition	New <i>classes</i> introduced over time; previous data not revisited.	Same <i>classes</i> but different input domains (scanner, site, acquisition).	Sequential <i>tasks</i> with distinct labels; task identity known.	New <i>modalities</i> (e.g., MRI→CT) introduced incrementally.	Data arrive sequentially with frequent updates and limited replay buffer.	Continuous data stream; each sample seen once (single-pass learning).
Task ID at Test	No	No	Yes	± Partial	No	No
Primary Objective	Maintain discrimination across old and new classes.	Achieve robustness to domain shift.	Learn task-specific representations with minimal interference.	Preserve shared semantics across modalities.	Adapt online to evolving data distributions with limited memory.	Continually adapt without replay; ensure stability in streaming setup.
Main Challenge	Catastrophic forgetting.	Representation drift.	Task separation vs. generalization.	Cross-modality alignment.	Stability–plasticity under memory constraints.	Extreme stability–plasticity under single-pass restriction.
Evaluation Protocol	Accuracy and forgetting across classes.	Cross-domain accuracy and robustness.	Average per-task accuracy.	Cross-modality transfer performance.	Online accuracy and buffer-based forgetting.	Streaming accuracy and adaptation curve.
Example (Medical Imaging)	Incremental disease classification.	MRI reconstruction across institutions.	Sequential organ segmentation.	MRI–CT incremental fusion.	Real-time ultrasound adaptation.	Continuous ECG or sensor learning without replay.

CT segmentation with adversarial preservation of task-invariant features [104]. More recent advances address severe class imbalance through logit-adjusted and margin-aware objectives [105], prototype-driven incremental segmentation without additional task heads [106], and endoscopic continual learning with replay and calibration (*EndoCIL*) [107]. Efficiency-oriented directions include dynamic model merging [108] and dual-learner consolidation for foundation model adaptation [109]. Moreover, frozen foundation encoders have demonstrated strong resilience for incremental dermatology classification without catastrophic forgetting [110].

Overall, modern medical CIL frameworks increasingly integrate replay, distillation, prototype-based consolidation, and calibration mechanisms to balance stability–plasticity trade-offs under scarce labels, domain heterogeneity, and extreme inter-class imbalance—offering a promising pathway toward scalable, privacy-compliant, and clinically adaptive incremental learning systems.

2) DOMAIN-INCREMENTAL LEARNING (DIL)

Domain-incremental learning (DIL) addresses distributional shifts across domains—such as scanners, imaging protocols, institutions, or staining variations—while maintaining a shared label space. Although originally studied under the broader continual learning (CL) paradigm, DIL directly aligns with the incremental learning (IL) framework in medical imaging, where adaptation occurs stage by stage as new domains or institutions are introduced without overlapping data access.

Early medical imaging studies anticipated DIL by targeting multi-center variability and domain robustness.

Karani et al. [111] introduced domain-specific batch normalization to enable cross-scanner MRI segmentation, while Chartsias et al. [112] proposed a factorized representation separating anatomical content from imaging contrast to promote domain generalization. Srivastava et al. [113] later formalized a low-resource DIL setting for chest X-rays using memory-efficient latent replay, paving the way for scalable domain-adaptive incremental learning.

Recent research expands DIL to handle complex distributional and modality-level shifts. Li et al. [114] introduced style-oriented replay and whitening for cardiac MRI segmentation (M&Ms dataset), while Kumari et al. [115] proposed privacy-aware generative latent replay for digital pathology under stain and organ variations. Wang et al. [116] presented a dual-branch disentanglement strategy for cross-domain chest X-ray learning, and Li et al. [117] combined shape-aware enrichment with multi-expert expansion for continual cardiac segmentation across institutions.

Hybrid DIL frameworks increasingly integrate transfer, federated, and uncertainty-aware learning paradigms. You et al. [118] designed site-agnostic encoders with site-specific decoders for incremental prostate MRI segmentation. Zhang et al. [89] proposed federated cross-incremental self-supervised learning (FedCSL), combining collaborative distillation with masked image modeling. Complementary works such as Huang et al. [119] surveyed incremental transfer across multi-center clinical datasets, while Hassan et al. [120] explored Bayesian regularization for cross-domain retinopathy screening.

Overall, medical DIL provides a practical incremental formulation of continual adaptation—enabling models to retain prior-domain knowledge while sequentially integrating

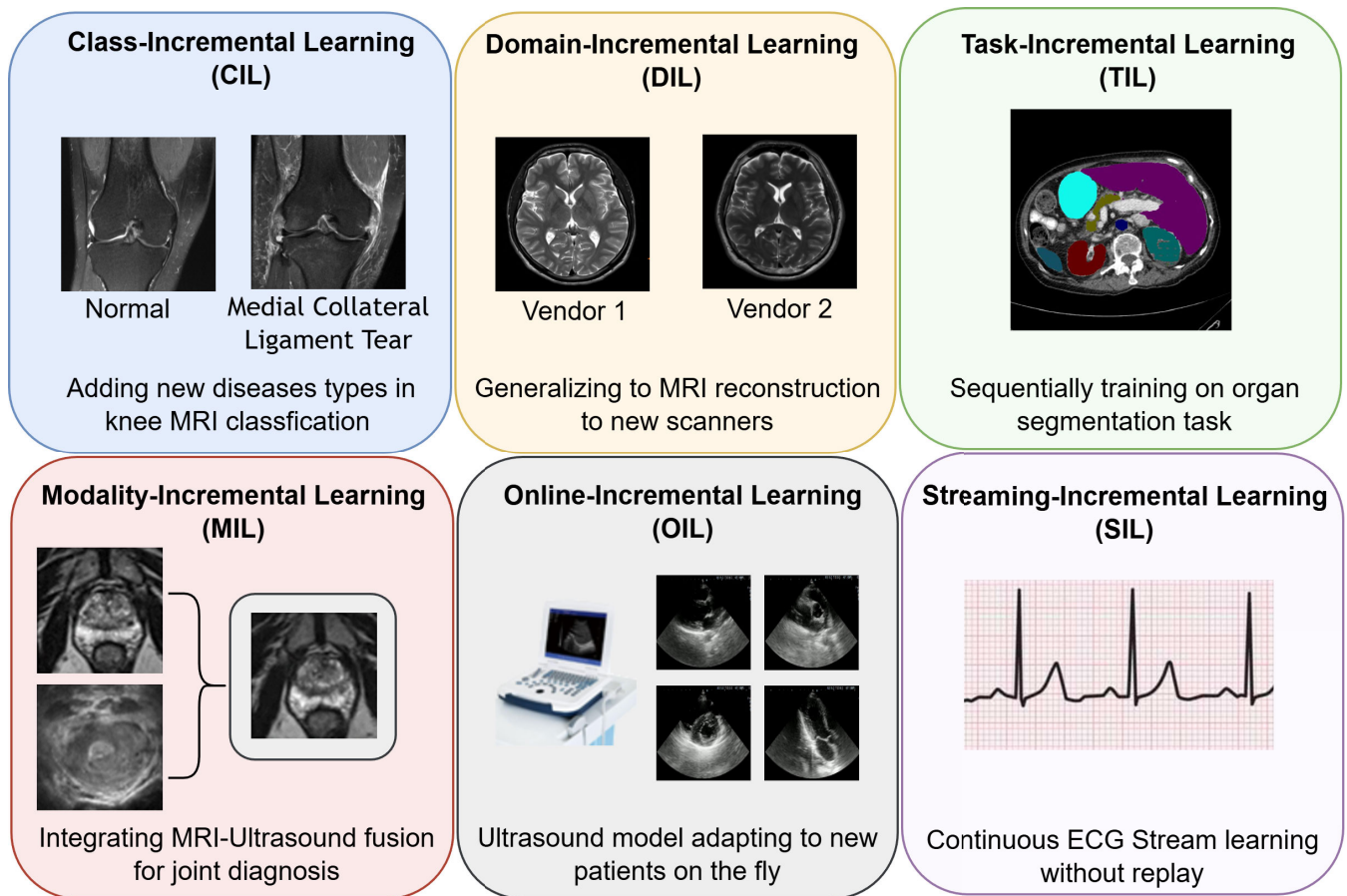


FIGURE 4. Comparative visualization of incremental learning scenarios in medical imaging. Class-incremental learning (CIL) introduces new disease categories; domain-incremental learning (DIL) adapts models to new scanners or institutions; task-incremental learning (TIL) sequentially learns organ segmentation tasks; modality-incremental learning (MIL) fuses cross-modality inputs (e.g., MRI-ultrasound); online-incremental learning (OIL) enables real-time model updates; and streaming-incremental learning (SIL) continuously learns from unbounded data streams such as ECG.

new sites, scanners, and protocols. Its success depends on disentangled representations, privacy-preserving replay, and federated or self-supervised objectives, collectively paving the way toward robust, adaptive, and clinically generalizable incremental learning across distributed healthcare environments.

3) TASK-INCREMENTAL LEARNING (TIL)

Task-incremental learning (TIL) assumes that the task identity is known during inference, typically activating task-specific heads or normalization parameters to minimize interference between tasks. Although originally introduced as part of the broader continual learning (CL) taxonomy [101], TIL naturally aligns with the incremental learning (IL) paradigm in medical imaging, where new diagnostic or anatomical tasks are introduced sequentially under controlled supervision.

This paradigm is particularly relevant to clinical workflows where objectives evolve stepwise—such as lesion detection → organ segmentation → disease classification. Foundational frameworks, including Progress & Compress [102], Learning without Forgetting (LwF) [10], and

context-dependent gating [121], established selective consolidation and parameter modulation mechanisms that later inspired medical TIL formulations.

Interpretable and biologically inspired TIL approaches have been explored in fetal heart anatomy analysis, where saliency-guided exemplar retention supports explainable continual adaptation [122]. Benchmarking efforts such as *LifeLonger* [123] standardized task, class, and cross-domain splits on MedMNIST, facilitating systematic evaluation across incremental setups. Parameter-efficient extensions—such as task-specific batch normalization with frozen shared kernels [124] and multi-scale, uncertainty-weighted distillation for 3D segmentation [86]—achieve strong stability-plasticity trade-offs under single-pass constraints. For organ- and tumor-level segmentation, modular class-specific heads and CLIP-based embeddings enable semantic transfer with minimal cross-task interference [63].

Overall, TIL provides a controlled and interpretable incremental formulation for multi-task medical imaging pipelines. Future directions point toward integrating task-conditioned adapters, federated continual training, and

modular head expansion to further enhance adaptability under heterogeneous clinical environments.

4) MODALITY-INCREMENTAL LEARNING (MIL)

Modality-incremental learning (MIL) represents a natural extension of the incremental learning (IL) paradigm to settings where new imaging modalities are introduced sequentially. Although originally discussed within the continual learning (CL) literature, MIL directly aligns with incremental adaptation in medical imaging—where new acquisition types (e.g., CT, MRI, PET, ultrasound) become available over time and must be integrated without retraining from scratch. Unlike joint multimodal training, MIL assumes that modalities arrive disjointly (e.g., CT $\rightarrow$ MRI $\rightarrow$ PET; fundus $\rightarrow$ OCT $\rightarrow$ OCT-A) and that models must maintain cross-modal consistency while expanding representational capacity.

In the general vision domain, Hegde et al. [125] formalized MIL through disjoint relevance mapping to mitigate inter-modal interference, while Song et al. [126] introduced *Harmony*, a framework for compatible feature modulation and cumulative modal bridging. Related multimodal continual learning works on dynamic transformers [70] and continual visual question answering (VQA) [127] further highlight the importance of adaptive expansion and compositional disentanglement across modalities.

In medical imaging, MIL has been explored to manage evolving modality availability and device heterogeneity. Morgado et al. [128] investigated device-induced modality variation in dermatology and demonstrated that replay-based methods effectively mitigate forgetting while maintaining classification accuracy. Yao et al. [129] proposed *RetCoP*, a continual vision–language pretraining framework that incrementally integrates fundus, OCT, and angiography modalities alongside textual clinical reports via image–text rehearsal and off-diagonal distillation.

Clinically, MIL is critical for progressive technology adoption—such as incorporating PET into MRI workflows—without requiring full model retraining. Persistent challenges include handling heterogeneous modality distributions, reconciling inconsistent annotation standards, and the lack of large-scale multimodal incremental benchmarks for evaluating cross-modality generalization. Future research directions include parameter-efficient multimodal adapters, foundation model tuning under sequential modality exposure, and privacy-preserving replay across imaging systems.

5) ONLINE AND STREAMING INCREMENTAL LEARNING (OIL/SIL)

Online and streaming incremental learning (OIL/SIL) extend the incremental learning (IL) paradigm to real-time and continuous data environments. Although originally developed within the continual learning (CL) literature, these paradigms align closely with incremental adaptation in

medical imaging and physiological monitoring, where data arrive sequentially or continuously without the possibility of full retraining. In this context, *online* learning focuses on one-pass updates from sequential inputs, whereas *streaming* learning generalizes this setup to temporally continuous, potentially unbounded data flows characterized by concept drift, delayed supervision, and strict memory or compute constraints [130], [131], [132].

a: FOUNDATIONS OF ONLINE INCREMENTAL LEARNING

Mai et al. [103] established a modern benchmark for online continual learning (OCL)—a precursor of OIL—by comparing replay-based (e.g., MIR [133]), regularization-based, and buffer-free strategies under diverse memory regimes. Buzzega et al. [134] proposed *Dark Experience Replay*, a task-free rehearsal framework leveraging self-distillation to mitigate catastrophic forgetting, while Parisi and Lomonaco [135] provided a biologically inspired perspective linking online adaptation to neural replay and synaptic plasticity. Subsequent work introduced optimization-level improvements such as adaptive orthogonal projection [136], memory summarization [137], and unit-sphere representation learning [138], which enhance robustness under blurry or evolving task boundaries. Harun et al. [139] presented *SIESTA*, a wake–sleep framework enabling rehearsal-free incremental adaptation on large-scale datasets such as ImageNet within strict compute budgets.

b: STREAMING INCREMENTAL LEARNING AND CONCEPT DRIFT

Beyond the standard online setup, streaming incremental learning (SIL) emphasizes continual adaptation under evolving data distributions driven by concept drift. Gama et al. [130] established the foundational taxonomy of drift adaptation, while Zhang et al. [131] proposed AOIL, an autoencoder-based drift detection framework with attention-driven latent updates. Prashanth et al. [132] demonstrated uncertainty-guided buffering and hybrid drift detection on clinical time series, achieving stable performance under both abrupt and gradual drifts.

c: APPLICATIONS IN MEDICAL IMAGING AND PHYSIOLOGY

In medical contexts, OIL and SIL underpin adaptive systems for dynamic imaging and physiological monitoring. Applications include annotation-efficient online active learning for histopathology and MRI [140], counterfactual incremental learning for streaming ultrasound [141], early online anomaly detection for vital signs [142], and privacy-preserving federated streaming across hospitals [143]. Together, these frameworks support deployment-ready clinical AI capable of incremental, real-time adaptation under drift, privacy, and latency constraints—representing a crucial bridge between incremental learning research and continuously evolving clinical environments.

D. APPLICATION DIMENSIONS (ORTHOGONAL AXES)

While previous sections categorize incremental learning paradigms by their algorithmic formulation (e.g., class-, domain-, task-, or modality-incremental), a complementary perspective focuses on the *application dimensions* that determine how these paradigms manifest across clinical contexts.

Although originally discussed within the continual learning (CL) literature, these application dimensions are directly aligned with the incremental learning (IL) paradigm, where adaptation occurs progressively under clinical and operational constraints. They bridge the gap between algorithmic design and real-world deployment by emphasizing the practical factors that govern how incremental systems evolve in medicine.

These dimensions—**task type**, **imaging modality**, **data access constraints**, and **challenge axis**—jointly characterize the interaction between incremental model design and practical deployment factors such as imaging diversity, annotation scarcity, and privacy limitations.

A comparative overview of these orthogonal dimensions is presented in **Table 4**, summarizing their definitions, subcategories, incremental manifestations, and the key challenges that arise when applying incremental learning in medical imaging.

1) TASK TYPE

Incremental learning (IL)—originally conceptualized under the continual learning (CL) paradigm—spans diverse medical imaging objectives, each posing unique stability–plasticity trade-offs. Across primary task categories such as classification, segmentation, detection, registration, and reconstruction, most studies emphasize class- and domain-incremental formulations to enable diagnostic scalability and lifelong adaptation of medical AI systems.

a: CLASSIFICATION

Class-incremental learning (CIL) remains the dominant paradigm, reflecting evolving diagnostic categories and imaging protocols. Notable frameworks such as EndoCIL [107] for endoscopic image classification and the foundation-based dermatology approach of Elkhayat et al. [110] demonstrate effective adaptation without full model retraining. Few-shot CIL variants (e.g., Fscil-EACA [144]) and multimodal extensions such as RetCoP [129] address data scarcity and cross-modal integration challenges.

b: SEGMENTATION

Incremental segmentation frameworks address scenarios where new organ or lesion labels appear over time. ACL-Seg [104], EM-based online segmentation [145], and transformer-driven CoMFormer [146] exemplify the progression from CNN-based replay mechanisms to adaptive

distillation frameworks, achieving improved retention of prior anatomical knowledge across incremental steps.

c: DETECTION, REGISTRATION, AND RECONSTRUCTION

Incremental detection frameworks (e.g., ILCM [147]) extend continual adaptation to emerging disease patterns and imaging protocols. Incremental registration methods such as WiNet [148] and semi-incremental cell-level alignment [149] enhance temporal and cross-organ consistency. Similarly, Nest-DGIL [150] demonstrates that incremental geometric learning improves reconstruction fidelity under dynamic imaging conditions.

Summary: Collectively, these efforts illustrate that incremental learning transcends traditional classification-focused setups—enabling lifelong adaptation across diagnostic, spatial, and temporal objectives in medical imaging, while maintaining a balance between stability and plasticity under evolving clinical data streams.

2) IMAGING MODALITY

Imaging modality constitutes a key orthogonal factor shaping incremental learning (IL) behavior in medical imaging. Although most modality-related analyses originate from the continual learning (CL) literature, the same challenges—catastrophic forgetting, replay feasibility, and domain drift—manifest distinctly across 2D, 3D, video-based, and gigapixel imaging data under incremental adaptation.

a: 2D AND RETINAL IMAGING

Early incremental frameworks primarily focused on 2D modalities such as fundus and chest X-ray imaging. RetCoP [129] extends this direction to multimodal retinal learning via image–text alignment, while incremental X-ray classification [147] demonstrates robust adaptation under low-data and streaming conditions.

b: 3D VOLUMETRIC IMAGING

Volumetric CT and MRI introduce high computational and memory demands for incremental learning. Multi-organ segmentation [151], [152] and cross-modality registration [153] frameworks maintain spatial coherence and reduce catastrophic forgetting across incremental updates. In MRI reconstruction, Nest-DGIL [150] exemplifies progressive fidelity improvement without full retraining, reflecting the emerging trend toward dynamic volumetric adaptation.

c: DIGITAL PATHOLOGY AND MICROSCOPY

Gigapixel whole-slide and microscopy images introduce domain heterogeneity from staining, scanner variability, and tissue diversity. Privacy-preserving domain-incremental approaches [115] and replay-based pathology classification [154] maintain stable feature embeddings across sequential updates, while incremental microscopy alignment [149] improves temporal and structural consistency in time-lapse imaging.

TABLE 4. Comparative overview of orthogonal application dimensions defining incremental learning in medical imaging.

Aspect	Task Type	Imaging Modality	Data Access Constraints	Challenge Axes
Definition	Defines the clinical or computational goal where incremental updates are performed.	Represents the imaging domain in which incremental learning is applied and adaptation occurs.	Describes how data accessibility and privacy regulations affect incremental model updates.	Captures external and systemic factors influencing stability–plasticity balance during incremental adaptation.
Subcategories	Classification, segmentation, detection, reconstruction.	MRI, CT, ultrasound, X-ray, fundus, histopathology.	Centralized, federated, decentralized, partially accessible, privacy-constrained.	Domain shift, limited supervision, class imbalance, memory constraint, real-time or streaming updates.
Incremental Manifestation	Sequential addition of new anatomical structures, disease categories, or diagnostic tasks.	Gradual integration of new modalities or cross-modal adaptation (e.g., MRI→CT).	Progressive inclusion of datasets from new institutions or cohorts under restricted access.	Continuous exposure to distributional drift, evolving annotations, or resource-constrained learning.
Primary Challenge	Task interference and catastrophic forgetting across sequential tasks.	Modality gap and feature inconsistency across imaging domains.	Heterogeneous and privacy-restricted data limiting shared incremental learning.	Maintaining stability–plasticity trade-off under imbalance, drift, or limited labels.
Evaluation Focus	Accuracy, Dice/F1 score, mAP, PSNR/SSIM (for reconstruction).	Cross-modality transferability and incremental adaptation metrics.	Federated or distributed performance retention, privacy–utility trade-off.	Forgetting rate, adaptation robustness, and incremental efficiency under evolving conditions.

d: ENDOSCOPIC AND VIDEO-BASED IMAGING

Video-based modalities, such as gastrointestinal endoscopy, naturally suit incremental learning due to continuous frame streams and evolving imaging conditions. EndoCIL [107] and efficient foundation-model adaptation techniques (e.g., PSF-SAM [155]) achieve strong generalization across centers and devices, mitigating domain drift in real-time settings.

Summary: Modality-aware incremental learning reveals that forgetting dynamics, representational stability, and replay feasibility vary substantially with imaging dimensionality and acquisition variability. These findings emphasize that modality-specific strategies—spanning from lightweight replay to dynamic feature modulation—are crucial for building robust, generalizable incremental systems across the medical imaging spectrum.

3) DATA ACCESS CONSTRAINTS

Incremental learning (IL) in medical imaging often operates under stringent data-access and regulatory constraints. Centralized data aggregation is rarely feasible due to privacy laws (e.g., GDPR, HIPAA) and institutional ownership of medical records. Accordingly, federated, decentralized, and privacy-preserving IL frameworks have been proposed to enable continual adaptation without direct data sharing. Federated incremental learning strategies leverage local updates with global aggregation while maintaining patient confidentiality, whereas decentralized approaches rely on peer-to-peer model synchronization. Emerging privacy-aware replay and secure gradient-sharing mechanisms further mitigate data leakage risks, ensuring compliance with clinical governance (see Section III-E). Overall, these data-access paradigms

determine how incremental models can be trained and deployed across multi-institutional or cross-border medical environments.

4) CHALLENGE AXES IN INCREMENTAL LEARNING

Beyond data access, incremental medical AI faces persistent challenge axes that affect the stability–plasticity balance and long-term retention. Key challenges include catastrophic forgetting, domain heterogeneity, class imbalance, annotation scarcity, and concept drift. Modern IL frameworks address these issues through generative replay, uncertainty calibration, adaptive bias correction, and foundation-based consolidation [109], [110]. Recent efforts also integrate self-supervised and uncertainty-aware objectives to handle label noise and domain shift more effectively, while memory-efficient strategies seek to balance computational cost with clinical scalability.

Together, these challenge axes define the core barriers to achieving reliable and generalizable incremental learning in medical imaging.

E. DATA ACCESS CONSTRAINTS

Data sharing and storage restrictions, particularly prevalent in clinical settings, impose major challenges for incremental adaptation in medical imaging. While most recent literature refers to these approaches under the broader term *continual learning (CL)*, their underlying principles are highly aligned with *incremental learning (IL)*—that is, adapting to new tasks or domains under non-stationary conditions without full retraining or centralized data aggregation. This subsection reviews recent advances that address such constraints

across four major dimensions: (i) privacy-preserving incremental learning, (ii) federated and decentralized frameworks, (iii) rehearsal-free and pseudo-replay strategies, and (iv) resource-efficient and lightweight update mechanisms.

1) PRIVACY-PRESERVING INCREMENTAL LEARNING

Privacy-preserving incremental learning (PPIL) focuses on maintaining model adaptability while complying with medical data protection and non-retention regulations. Verma et al. [37] conducted a comparative analysis of PPIL algorithms for medical image classification, demonstrating that exemplar-free Bayesian and elastic regularization methods outperform joint retraining under privacy constraints. Yoo et al. [156] introduced Review Learning (RevL), a multi-institutional framework that incrementally refines models using self-generated pseudo-samples instead of historical patient data. In digital pathology, Kumari et al. [115] proposed a domain-incremental model that preserves privacy while adapting to staining or scanner shifts through latent generative replay. Zhu et al. [157] and Khalid et al. [158] provided broad surveys of privacy-preserving techniques such as encryption, differential privacy (DP), homomorphic encryption, and federated learning, which form the foundation for secure incremental adaptation in medical imaging. Earlier DP-based IL studies [159], [160] further explored the trade-off between privacy budgets and task performance, offering key insights for privacy-aware incremental design.

2) FEDERATED AND DECENTRALIZED FRAMEWORKS

Federated incremental learning (FIL) extends IL/CL paradigms to multi-center settings, where models are collaboratively updated without raw data sharing. Yang [143] investigated FedEWC and FedLwF for class-incremental medical imaging, revealing that elastic weight consolidation mitigates forgetting more effectively under inter-site heterogeneity. Kapoor et al. [161] proposed FedCL-Ensemble, integrating ensemble transfer learning within a federated incremental setup for Alzheimer's MRI classification, improving stability and privacy compliance. Zhou and Wang [162] presented a cross-paradigm fusion of federated and incremental learning using an MLP-Mixer backbone for thoracic infection diagnosis. Mennour [163] introduced a decentralized federated approach supporting sequential healthcare data updates, while Masuyama et al. [164] developed a privacy-preserving federated clustering model via Adaptive Resonance Theory. Complementary works such as Koutsoubis et al. [165] and Hamed et al. [166] emphasized ongoing challenges related to inter-client imbalance, trust calibration, and decentralized optimization in medical federated incremental systems.

3) REHEARSAL-FREE AND PSEUDO-REPLAY STRATEGIES

To address privacy and memory limitations, recent research has emphasized data-free and generative replay approaches that avoid explicit storage of prior samples. Zhang et al. [167]

proposed a data-free incremental learning framework in federated environments using diffusion-based prototypes to synthesize prior distributions. Babakniya et al. [168] introduced a federated *class-incremental* approach that combines generative replay with exemplar-free adaptation for vision tasks. Wu et al. [169] developed a contrastive reversion framework for data-free IL, reconstructing feature-level information without historical exemplars. In whole-slide image (WSI) analysis, Kumari et al. [170] proposed Attention-based Generative Latent Replay (AGLR), which replays Gaussian mixture representations to preserve prior knowledge under privacy constraints. Byun et al. [51], [171] extended this line of work with diffusion- and prompt-based generative replay frameworks, enhancing robustness for diabetic retinopathy and dermoscopic image classification. Qu et al. [172] explored collaborative generative replay for distributed medical imaging, illustrating effective alignment across heterogeneous federated clients.

4) RESOURCE-EFFICIENT AND LIGHTWEIGHT UPDATES

In resource-constrained or edge medical environments, incremental learning must balance computational efficiency with adaptability. Huang et al. [173] proposed an edge-based micro-training framework for privacy-aware IL, where less than 2% of model parameters are updated per iteration. Öet al. [174] introduced Privacy-Aware Lifelong Learning (PALL), leveraging sparse subnetwork activation and episodic memory for efficient continual adaptation. Li et al. [175] reviewed IL methods for physiological signal data, emphasizing interpretability and low-energy deployment in clinical environments. Iqbal et al. [176] combined federated and incremental paradigms into a “Wisdom Learning” framework, improving fairness and robustness under non-IID and biased updates. Smith et al. [177], [178] explored both incremental and continual visual-language learning, proposing parameter-efficient, data-free strategies aligned with privacy-aware and scalable medical deployment.

5) SUMMARY

Collectively, these studies demonstrate how incremental and continual learning frameworks are evolving to address the realities of medical data governance. Privacy-preserving and federated approaches enable cross-institutional adaptation without compromising confidentiality, while data-free replay and lightweight updates reduce storage and computational overhead. Together, these advances represent foundational steps toward translating incremental learning from theoretical research into clinically practical and regulatory-compliant systems.

F. CHALLENGE AXES

Incremental and continual learning in medical imaging are shaped by multiple forms of distributional and structural non-stationarity. These “challenge axes” capture how medical

data, labels, and clinical workflows evolve over time—often violating the independent and identically distributed (i.i.d.) assumptions underlying conventional deep learning [20], [101]. Understanding these axes is crucial for designing models that not only retain prior knowledge but also adapt stably to new domains, modalities, and diagnostic objectives [19], [115]. The following subsections outline the primary sources of these challenges and their implications for incremental and continual adaptation.

1) DOMAIN AND MODALITY SHIFT

Medical imaging data are inherently heterogeneous across institutions, scanners, acquisition protocols, and patient populations. Domain shifts arise from variations in imaging parameters (e.g., field strength, vendor calibration, contrast dosage) or modality-specific characteristics (e.g., MRI vs. CT vs. histopathology) [111], [114], [116]. Such distributional drift leads to catastrophic forgetting when models are sequentially trained on non-overlapping domain streams, particularly in incremental or federated setups [89], [118]. In federated frameworks, domain shift manifests as inter-site heterogeneity in texture, intensity, or resolution distributions without shared data access [119]. Mitigation strategies typically involve domain-invariant representation learning, contrastive alignment, or generative replay to stabilize cross-domain adaptation [115], [117].

2) LABEL AND TASK DRIFT

Label drift occurs when diagnostic or segmentation categories evolve, merge, or reappear under new clinical taxonomies, while task drift extends this challenge to entirely new prediction objectives (e.g., transitioning from organ segmentation to lesion classification) [122], [123]. Such shifts are common due to evolving diagnostic standards and dataset-specific labeling conventions [86], [124]. Unlike static benchmarks, incremental systems must incorporate new label semantics without erasing prior clinical knowledge. Regularization-based stability constraints, meta-learning, and dynamic architectural expansion have been explored to preserve task boundaries, while task-agnostic memory and representation replay maintain old-task fidelity [102], [107].

3) CLASS IMBALANCE AND DATA HETEROGENEITY

Class imbalance is especially severe in medical imaging, where rare or pathological cases are underrepresented. In incremental settings, this imbalance worsens as rare classes appear late or sporadically in the training stream [105], [109]. The model may overfit to dominant recent classes, accelerating forgetting of earlier minority categories. Patient-level heterogeneity—arising from demographic, pathological, or acquisition variability—further amplifies this challenge. Recent approaches integrate adaptive reweighting, rehearsal balancing, uncertainty-driven sampling, and pseudo-replay to mitigate imbalance during sequential updates [108], [151].

4) ANNOTATION SCARCITY AND SUPERVISION GAPS

Label acquisition in medical imaging is costly and often limited to partial, weak, or noisy annotations [104], [145]. Incremental learning systems must therefore adapt to new tasks under limited supervision, frequently leveraging pseudo-labeling, self-supervision, or consistency-based objectives [119], [129]. Annotation scarcity also hampers replay and evaluation, as earlier data may lack full annotations for newly introduced diagnostic categories. To address these gaps, self-distillation, representation replay, and multi-modal contrastive pretraining have been employed to enhance data efficiency and stability [89], [117].

5) EVALUATION AND BENCHMARKING INSTABILITY

Unlike fixed computer vision benchmarks, medical datasets evolve dynamically due to updated acquisition protocols, changing inclusion criteria, and limited data sharing. This temporal instability complicates reproducibility and fairness in incremental evaluation [107], [123]. Recent initiatives propose standardized evaluation protocols—such as sequential domain shifts, open-set configurations, and multi-institutional data streams—to improve comparability across medical IL/CL studies [103], [179].

6) SUMMARY

These challenge axes define the operational landscape of incremental and continual learning in medical imaging. They reveal that medical continual adaptation extends beyond knowledge retention: it requires navigating evolving domains, shifting label semantics, and supervision gaps within real-world clinical workflows. Addressing these axes jointly demands hybrid strategies that balance stability, plasticity, and privacy under realistic data-governance constraints. Collectively, these insights underscore why incremental and continual paradigms are indispensable for clinically viable lifelong learning in medical imaging.

IV. EVALUATION IN INCREMENTAL MEDICAL LEARNING

Evaluation in incremental medical learning (IL) systematically assesses how effectively a model adapts to new clinical tasks while preserving diagnostic knowledge from previously learned ones. Unlike open-set or lifelong paradigms, IL assumes a sequential stream of related tasks or distributions—such as new disease categories, imaging modalities, or acquisition centers—where the model has access only to the current task's data. This setup mirrors realistic clinical workflows, where past data are unavailable due to privacy, storage, or governance constraints.

A. EVALUATION PROTOCOLS

1) INCREMENTAL TASK FORMULATION

Medical IL experiments are typically organized into a sequence of tasks $\{T_1, T_2, \dots, T_n\}$, where each task introduces new diagnostic categories, imaging domains, or acquisition variations. The model learns these tasks consecutively

without revisiting earlier data, reflecting the longitudinal and privacy-restricted nature of clinical data streams. This formulation allows systematic assessment of adaptability (plasticity) and retention (stability) under non-stationary and privacy-limited conditions.

2) EVALUATION SETUP AND DATA PARTITIONING

Datasets are partitioned into a series of increments following one of the common IL protocols:

- **Class-Incremental:** new disease or anatomical categories are introduced over time (e.g., additional lesion or tumor classes).
- **Domain-Incremental:** data from new domains or institutions appear sequentially (e.g., MRI scanners, staining variations, or hospital sites).
- **Task-Incremental:** each increment represents a distinct prediction objective (e.g., segmentation $\rightarrow$ classification).

The protocol choice depends on dataset structure and target application, ensuring that evaluation captures realistic continual updates in clinical deployment.

3) EVALUATION PHASES

After training on each task T_t , the model is evaluated on all previously encountered tasks $\{T_1, \dots, T_t\}$ to measure:

- 1) **Current Task Accuracy (CTA)** — performance on newly introduced data;
- 2) **Backward Transfer (BWT)** — change in performance on previous tasks (knowledge retention);
- 3) **Forward Transfer (FWT)** — facilitation of future learning through prior representations.

Aggregating these results across all tasks yields the *Average Incremental Accuracy (AIA)*, reflecting cumulative model stability and adaptability.

4) REFERENCE BASELINES

To ensure fair comparison, IL models are typically benchmarked against:

- a *Joint model*, trained on all data simultaneously (upper bound);
- a *Sequential fine-tuning baseline*, updated without forgetting mitigation (lower bound);
- representative IL strategies such as replay-based, regularization-based, or generative replay methods.

These baselines capture trade-offs between adaptability, memory efficiency, and stability under sequential updates.

5) CLINICAL REALISM AND LONGITUDINAL VALIDATION

Recent studies increasingly emphasize realism by structuring tasks along temporal or institutional axes—such as multi-center histopathology studies or longitudinal MRI acquisitions. Here, evaluation focuses not only on predictive accuracy but also on robustness to domain drift, temporal evolution, and data-access constraints, ensuring clinically sustainable and privacy-compliant model adaptation.

B. EVALUATION METRICS

Assessing performance in incremental medical learning requires evaluating both predictive accuracy and knowledge retention across evolving tasks. Unlike conventional single-stage learning, IL metrics capture how effectively a model integrates new information while mitigating catastrophic forgetting. While many of these metrics were originally proposed in the continual learning (CL) literature, they are now widely adopted in medical IL to quantify stability, adaptability, and long-term diagnostic reliability. Table 5 summarizes the primary metrics used in incremental medical imaging frameworks.

1) RATIONALE FOR METRIC SELECTION

Incremental medical imaging requires simultaneous assessment of three complementary aspects: (i) task-level diagnostic performance, (ii) knowledge retention over time, and (iii) safe adaptation to newly arriving data. No single metric is sufficient to capture these dimensions. We therefore adopt a combination of IL-specific metrics (e.g., AIA, FR, BWT/FWT, Stability–Plasticity) to quantify learning dynamics and forgetting behavior, together with task-specific clinical metrics (e.g., Dice, IoU, AP) that reflect spatial and diagnostic accuracy. In addition, reliability-oriented measures such as Expected Calibration Error (ECE) and longitudinal performance drift are included because clinically deployed systems must provide stable and well-calibrated predictions under evolving data distributions. This selection follows common practice in both continual learning and medical imaging literature, and is motivated by the need to balance accuracy, retention, and clinical safety in incremental deployment scenarios.

2) AVERAGE INCREMENTAL ACCURACY (AIA)

$$AIA = \frac{1}{N} \sum_{i=1}^N R_{N,i}$$

Quantifies mean performance across all tasks after the final training stage, reflecting cumulative diagnostic stability.

3) FORGETTING RATE (FR)

$$FR = \frac{1}{N-1} \sum_{i=1}^{N-1} \max_{j < N} (R_{j,i} - R_{N,i})$$

Measures the degree of catastrophic forgetting; lower FR indicates better retention of prior diagnostic knowledge.

4) BACKWARD AND FORWARD TRANSFER (BWT/FWT)

$$BWT = \frac{1}{N-1} \sum_{i=1}^{N-1} (R_{N,i} - R_{i,i}),$$

$$FWT = \frac{1}{N-1} \sum_{i=2}^N (R_{i,i} - b_i)$$

Together, they quantify interference and positive transfer across tasks—crucial for evaluating cross-domain or cross-modality adaptation in medical IL.

5) STABILITY–PLASTICITY RATIO

$$SP = \frac{A_{old}}{A_{new}}$$

Represents the balance between knowledge retention (stability) and adaptability (plasticity); an optimal ratio near 1 suggests balanced continual adaptation.

6) TASK-SPECIFIC AND CLINICAL METRICS

Complementary domain metrics extend IL evaluation to classification, segmentation, and detection tasks:

- **Area Under the ROC Curve (AUROC):** threshold-independent diagnostic discrimination for classification tasks;
- **Dice Similarity Coefficient (DSC):** region overlap in segmentation tasks;
- **Intersection over Union (IoU):** multi-organ or lesion-level segmentation accuracy;
- **Average Precision (AP):** detection reliability across confidence thresholds.

7) LONGITUDINAL CONSISTENCY AND DRIFT ANALYSIS

For temporal or multi-center datasets, performance drift over time is measured as:

$$Drift = |A_t - A_{t-1}|$$

This captures temporal robustness and domain stability under evolving data distributions.

a: SUMMARY

Together, these protocols and metrics form a comprehensive evaluation framework for incremental medical learning, integrating diagnostic accuracy, retention stability, and adaptability under clinically realistic, privacy-aware, and temporally evolving conditions.

C. BENCHMARK DATASETS IN INCREMENTAL MEDICAL IMAGING

In addition to evaluation protocols and metrics, a major challenge in incremental medical learning (IL) is the lack of standardized benchmarks that reflect clinically realistic non-stationarity. Unlike natural-image continual learning benchmarks (e.g., Split-MNIST or CIFAR), medical imaging streams are often shaped by (i) incremental emergence of new disease categories, (ii) institutional domain shifts caused by different scanners and acquisition protocols, and (iii) temporally evolving cohorts with changing inclusion criteria, label definitions, and annotation schemas. Moreover, unlike curated computer vision benchmarks, medical IL frequently involves concurrent drift in domain, label distribution, and supervision quality, which complicates

reproducibility and fair comparison. To facilitate clinically meaningful and reproducible evaluation, Table 6 summarizes representative benchmark datasets used in incremental medical imaging and highlights their typical incremental settings (class-, domain-, or modality-incremental), task types, and evaluation focus.

These benchmarks are typically evaluated under sequential training protocols (Section IV) using both IL-specific metrics (e.g., AIA, FR, BWT/FWT) and task-specific clinical metrics (e.g., Dice, mAP, AUROC), enabling clinically realistic comparison of retention and adaptation across diverse incremental axes.

V. OPEN CHALLENGES AND RESEARCH DIRECTIONS

Despite significant progress in incremental learning (IL) for medical imaging, several open challenges continue to limit its reliable adoption in clinical workflows. These limitations arise from data scarcity, model instability, privacy regulations, and the lack of unified evaluation protocols. This section outlines key unresolved issues and emerging research directions toward developing scalable, privacy-preserving, and clinically trustworthy incremental medical learning systems.

A. DATA AND ANNOTATION CONSTRAINTS

Incremental medical learning remains constrained by the scarcity, heterogeneity, and privacy restrictions of annotated medical data. Unlike general vision datasets, clinical collections are typically small, imbalanced, and acquired under non-stationary conditions across institutions, scanners, and time. Such variability amplifies domain shift and catastrophic forgetting.

Future research should emphasize:

- **Data-efficient replay:** Employ diffusion or text-conditioned generative replay to synthesize realistic, task-specific samples without accessing original patient data.
- **Self-supervised and active incremental learning:** Leverage unlabeled or weakly annotated data streams with uncertainty-driven querying for expert labeling efficiency.
- **Synthetic domain expansion:** Utilize GAN-, VAE-, or diffusion-based augmentation for continual domain expansion across scanners and protocols.

1) STABILITY–PLASTICITY DILEMMA

Balancing knowledge retention (stability) and adaptability (plasticity) remains the central challenge in IL. Regularization-based methods (e.g., EWC, LwF) may overconstrain adaptation, whereas replay-based approaches raise privacy and memory concerns.

Promising directions include:

- **Dynamic regularization:** Introduce adaptive task-weighting and memory-aware constraints to stabilize learning across stages.

TABLE 5. Summary of evaluation metrics in incremental medical learning.

Metric	Purpose	Definition / Measure	Application in Medical IL
Average Incremental Accuracy (AIA)	Evaluates cumulative performance across all tasks	$AIA = \frac{1}{T} \sum_{i=1}^T R_{T,i}$	Measures final diagnostic accuracy after sequential updates across disease or modality streams
Forgetting Rate (FR)	Quantifies catastrophic forgetting on earlier tasks	$FR = \frac{1}{T-1} \sum_{i=1}^{T-1} \max_{t < T} (R_{t,i} - R_{T,i})$	Evaluates retention of prior anatomical or diagnostic patterns under incremental adaptation
Backward Transfer (BWT)	Measures effect of new learning on old tasks	$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i})$	Identifies positive or negative transfer across sequential radiology or histopathology tasks
Forward Transfer (FWT)	Quantifies how prior knowledge aids new learning	$FWT = \frac{1}{T-1} \sum_{i=2}^T (R_{i,i} - b_i)$	Evaluates cross-modality representation reuse (e.g., MRI → CT)
Stability–Plasticity (SP) Ratio	Balances knowledge retention and adaptability	$SP = \frac{A_{old}}{A_{new}}$	Analyzes trade-offs between preserving prior knowledge and integrating new disease categories
Retention Ratio (RR)	Measures preservation of previous task knowledge	$RR = \frac{R_{T,i}}{R_{i,i}}$	Used for assessing robustness under domain or scanner shift
Dice Similarity Coefficient (DSC)	Measures region overlap in segmentation	$DSC = \frac{2 P \cap G }{ P + G }$	Quantifies spatial consistency for incremental lesion or organ segmentation
Intersection over Union (IoU)	Evaluates segmentation accuracy	$IoU = \frac{ P \cap G }{ P \cup G }$	Used for assessing incremental learning in multi-organ segmentation
Average Precision (AP)	Evaluates detection accuracy	$AP = \int_0^1 p(r) dr$	Applied to incremental lesion or region detection tasks
Performance Drift (PD)	Captures performance variation across increments	$PD = A_t - A_{t-1} $	Reflects longitudinal stability in temporally evolving datasets
Expected Calibration Error (ECE)	Evaluates predictive reliability	$ECE = \sum_{m=1}^M \frac{ B_m }{n} acc(B_m) - conf(B_m) $	Assesses uncertainty calibration during continual clinical adaptation

- **Parameter-efficient adaptation:** Apply low-rank adaptation (LoRA), adapter tuning, or selective subnet updates for modular and efficient fine-tuning.
- **Neuro-inspired consolidation:** Explore biologically motivated mechanisms such as synaptic consolidation or meta-continual learning to enhance long-term representation stability.

2) FOUNDATION AND MULTIMODAL MODEL ADAPTATION

Large pretrained foundation models—such as vision-language or multimodal medical encoders—pose new challenges for incremental adaptation. Continually fine-tuning these large-scale models can lead to catastrophic forgetting across modalities or prompt shifts, while full retraining remains computationally infeasible. Emerging research explores parameter-efficient continual tuning, cross-modal replay, and prompt-based adaptation to maintain stability across evolving modalities and clinical tasks. Future IL systems must reconcile foundation model adaptation with continual, data-efficient, and privacy-aware learning in multimodal medical contexts.

3) FEDERATED AND DECENTRALIZED LEARNING BARRIERS

Federated and decentralized IL frameworks aim to enable continual adaptation across clinical institutions without data sharing. However, practical deployment is hindered by non-IID data, asynchronous participation, and communication constraints. Although differential privacy and secure aggregation protocols ensure confidentiality, they often degrade model accuracy and convergence.

Future research should target:

- **Communication-efficient updates:** Employ adaptive quantization, gradient sparsification, or selective parameter sharing to minimize overhead.
- **Privacy-preserving continual adaptation:** Integrate differential privacy, generative replay, and secure computation within hybrid federated architectures.
- **Decentralized lifelong learning:** Explore blockchain or peer-to-peer mechanisms to enable transparent, verifiable model evolution without central coordination.

4) EVALUATION AND BENCHMARK GAPS

Current IL benchmarks—often adapted from general continual learning (e.g., Split-MNIST, CIFAR-100)—fail to represent the temporal, heterogeneous, and privacy-governed nature of medical data. Most medical IL studies still rely on static, limited datasets, hindering fair assessment of robustness, generalization, or clinical realism.

Future work should prioritize:

- **Standardized evaluation protocols:** Harmonize metrics across accuracy, forgetting, calibration, and privacy dimensions.
- **Longitudinal multi-center benchmarks:** Develop datasets capturing domain drift, label evolution, and modality variation across time and institutions.
- **Reproducible continual frameworks:** Establish open pipelines for model lifecycle tracking, data versioning, and regulatory-compliant continual evaluation.

5) UNCERTAINTY AND TRUST CALIBRATION

Incremental learning systems often exhibit calibration drift as tasks evolve, undermining clinical reliability and decision

TABLE 6. Representative benchmark datasets used in incremental medical imaging (ILMI). CIL: class-incremental; DIL: domain-incremental; TIL: task-incremental; MIL: modality-incremental.

Dataset	Modality	Task	IL Setting	Metric(s)	Clinical incremental axis
MedMNISTv2 [109], [123]	Multi-modal (2D)	Classification	CIL / TIL	Acc, AUC, F1	Standardized low-complexity benchmark for disease class- and task-incremental learning.
ISIC 2019, HAM10000, Dermofit, Derm7pt [85], [105], [110]	Dermoscopy	Classification	CIL	Acc, AUC, Macro-F1	Long-tailed dermatology benchmarks emphasizing imbalance-driven forgetting.
EyePACS, RFMiD, KDRDD [105], [119]	Fundus	Classification	CIL / DIL	AUC, Acc	Retinal screening under class and cross-center domain shift.
MM-Retinal, FFA-IR [129]	Fundus (multi-modal)	Representation learning	MIL	Forgetting, AUC	Modality-incremental vision-language pretraining for unified retinal representations.
Duke-I/II/III, Zhang, Rabbani, BIOMISA [120]	OCT/Fundus	Classification	DIL	AUC, F1	Cross-device domain-incremental retinopathy screening.
ChestX-ray8, PadChest, CheXpert(+), ChestX-ray14, MIMIC-CXR [113], [116]	X-ray	Classification	DIL	AUC, Acc	Canonical domain-incremental radiology benchmarks with strong institution shift.
ACDC 2017, M&Ms [112], [114], [117]	Cardiac MRI	Segmentation	DIL	Dice, IoU	Multi-vendor cardiac MRI segmentation under protocol drift.
HCP, ADNI, ABIDE, IXI [111]	Brain MRI	Segmentation	DIL	Dice, IoU	Lifelong MRI segmentation across scanner and protocol shifts.
NCI-ISBI13, I2CVB, PROMISE12 [118]	Prostate MRI	Segmentation	DIL	Dice, HD95	Sequential multi-site prostate segmentation with non-shared data.
AAPM Thoracic Dataset [104]	CT	Segmentation	CIL	Dice, IoU	Replay-free class-incremental organ segmentation under privacy constraints.
BTCV, BraTS2015 [86], [106]	LiTS, CT / MRI (3D) [63],	Segmentation	CIL	Dice, IoU, HD95	Class-incremental 3D organ and lesion segmentation with high computational and memory demand.
PATH16, SKIN8 [108]	Histopathology (patch)	Classification	CIL	Acc, AUC	Patch-level class-incremental histopathology under resource constraints.
BRACS, CRC [115]	Histopathology (WSI)	Classification	DIL	AUC, F1	Privacy-aware domain-incremental learning under stain and multi-center shift.

confidence. Recent research advocates continual calibration mechanisms, Bayesian uncertainty propagation, and post-hoc trust correction to ensure reliability across sequential updates. Future work should integrate uncertainty quantification and trust calibration as core evaluation dimensions for incremental medical learning, ensuring stable confidence and interpretability over time.

6) ETHICAL, LEGAL, AND CLINICAL INTEGRATION

Deploying incremental learning systems in healthcare introduces unique ethical and regulatory challenges.

Continual model evolution can conflict with static regulatory approvals and complicate accountability, interpretability, and traceability.

Future research should focus on:

- **Transparent and auditable adaptation:** Embed continual learning within explainability and audit frameworks to ensure interpretability and traceability.
- **Uncertainty-aware validation:** Design continual validation protocols that monitor calibration, confidence, and decision stability over time.

- **Regulatory and privacy alignment:** Develop privacy-by-design IL architectures compliant with GDPR, HIPAA, and other healthcare regulations.

a: SUMMARY

In summary, advancing incremental medical learning demands a shift toward data-efficient, privacy-preserving, and ethically responsible systems that can evolve safely over time. Addressing these open challenges—including foundation model adaptation, uncertainty calibration, and clinical integration—will enable future AI systems to function as continuously improving, transparent, and trustworthy decision-support tools, bridging algorithmic innovation with real-world healthcare impact.

VI. FUTURE PERSPECTIVES

Incremental learning (IL) in medical imaging is evolving from isolated experimental studies toward clinically grounded, privacy-aware, and multimodal systems. Future research will likely emphasize scalability, generalization, and trustworthiness—bridging adaptive machine intelligence with real-world clinical deployment. This section outlines the emerging directions that are expected to define the next generation of incremental medical AI.

A. FOUNDATION MODELS AND CONTINUAL ADAPTATION

Recent advances in large-scale foundation models (e.g., SAM, MedSAM, CLIP, BioViL, and BioGPT) have demonstrated remarkable generalization across imaging modalities and clinical tasks. Integrating incremental learning principles into these models represents a critical step toward continuously adaptive, domain-robust medical intelligence capable of evolving with clinical data streams.

Key future directions include:

- **Parameter-efficient continual tuning:** Leveraging adapters, LoRA, or low-rank updates to enable scalable and memory-efficient lifelong fine-tuning of foundation models.
- **Task-agnostic continual pretraining:** Developing streaming-based self-supervised learning strategies for progressive knowledge accumulation across diverse medical domains.
- **Cross-modal continual adaptation:** Preserving joint alignment between image, report, and text embeddings to support dynamic multimodal reasoning and temporal consistency.

1) MULTIMODAL AND CROSS-INSTITUTIONAL INCREMENTAL LEARNING

Next-generation clinical AI systems must process multimodal data streams—combining images, clinical notes, and metadata—while adapting across hospitals, scanners, and populations. Achieving such generalization requires robust domain alignment under strict privacy and annotation constraints.

Promising research avenues include:

- **Multimodal fusion under continual updates:** Integrating CLIP-like image–text frameworks with incremental adaptation to preserve semantic and clinical consistency across modalities.
- **Cross-institutional incremental learning:** Enabling domain-agnostic adaptation that handles institutional variability without direct data sharing.
- **Lifelong personalization:** On-device or patient-specific adaptation enabling longitudinal model refinement while maintaining privacy guarantees.

2) GENERATIVE REPLAY AND SYNTHETIC MEMORY

Generative replay has emerged as a key strategy for mitigating catastrophic forgetting without storing sensitive patient data. Future IL frameworks may leverage diffusion and text-to-image models as privacy-safe generative memories capable of reconstructing past knowledge distributions and supporting continual adaptation.

Emerging directions include:

- **Text-conditioned diffusion replay:** Using vision–language models to generate representative samples from textual task descriptions or radiology reports.
- **Cross-task synthesis:** Jointly generating heterogeneous modalities to support continual domain adaptation and class balancing.
- **Hybrid replay–regularization frameworks:** Combining generative memory with knowledge-consolidation objectives for improved stability and adaptability.

3) TRUSTWORTHY, TRANSPARENT, AND EXPLAINABLE CONTINUAL AI

For clinical deployment, incremental models must be interpretable, calibrated, and auditable as they evolve over time. Ensuring transparency and reliability in continually adapting systems is critical for clinical safety and trust.

Future research should explore:

- **Uncertainty-aware continual inference:** Incorporating Bayesian or calibration-based approaches to quantify predictive reliability under evolving data distributions.
- **Explainable memory consolidation:** Visualizing how previously learned representations influence current predictions to enhance clinician interpretability.
- **Ethical and transparent adaptation:** Embedding continual learning within auditable pipelines aligned with regulatory and fairness principles.

4) TOWARD LIFELONG AND SELF-EVOLVING MEDICAL AI

Ultimately, incremental learning paves the way toward lifelong, self-evolving medical AI systems that can safely adapt to new clinical conditions and populations. By integrating foundation models, generative replay, federated learning, and uncertainty estimation, future frameworks will enable continuous model evolution without compromising privacy,

safety, or interpretability. Such systems will redefine the paradigm of clinical artificial intelligence—from static, one-time-trained models to adaptive, resilient, and trustworthy lifelong learners capable of evolving with medicine itself.

VII. CONCLUSION

Incremental learning (IL) has emerged as a transformative paradigm for developing adaptive, privacy-preserving, and resilient artificial intelligence (AI) systems in medical imaging. Unlike conventional static models trained on fixed datasets, IL frameworks allow AI systems to evolve as new imaging modalities, diagnostic categories, or clinical data streams become available. This adaptability is crucial for real-world deployment, where domain shifts, institutional variability, and stringent privacy regulations continually challenge model robustness and generalization.

In this review, we presented a comprehensive synthesis of incremental learning methodologies in medical imaging—covering privacy-preserving adaptation, federated and decentralized collaboration, rehearsal-free and generative replay strategies, as well as evaluation protocols and metrics for continual assessment. Our analysis shows that while recent advances have achieved notable progress, several key challenges remain unresolved, including catastrophic forgetting, federated heterogeneity, and the absence of standardized benchmarks and evaluation frameworks tailored to clinical data dynamics.

Future research should focus on three strategic directions:

- Establishing standardized, longitudinal benchmarks and continual evaluation pipelines for privacy-constrained medical data;
- Integrating incremental learning with large-scale foundation and multimodal models to enhance cross-domain adaptability; and
- Advancing interpretability, fairness, and clinical trust through uncertainty-aware and auditable continual AI frameworks.

In the long term, incremental medical learning represents a pathway toward lifelong, self-improving AI systems that can continuously integrate new knowledge while retaining prior expertise. Such systems have the potential to enable adaptive diagnostic tools, dynamic screening pipelines, and personalized treatment monitoring—ultimately transforming medical intelligence from static diagnostic models into adaptive, collaborative, and trustworthy lifelong learners that evolve alongside clinical practice.

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FATMA HARMAN (Associate Member, IEEE) received the B.Sc. degree in electrical and electronics engineering from Balıkesir University, Türkiye, in 2011, the M.Sc. degree in electrical and electronics engineering from Manisa Celal Bayar University, Türkiye, in 2016, and the Ph.D. degree in electrical and electronics engineering from Dokuz Eylül University, İzmir, Türkiye. She has authored several articles on deep learning-based MRI reconstruction and diagnostic image analysis.

Her research interests include machine learning, medical image analysis, magnetic resonance imaging (MRI) reconstruction, and open set recognition in medical imaging.

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